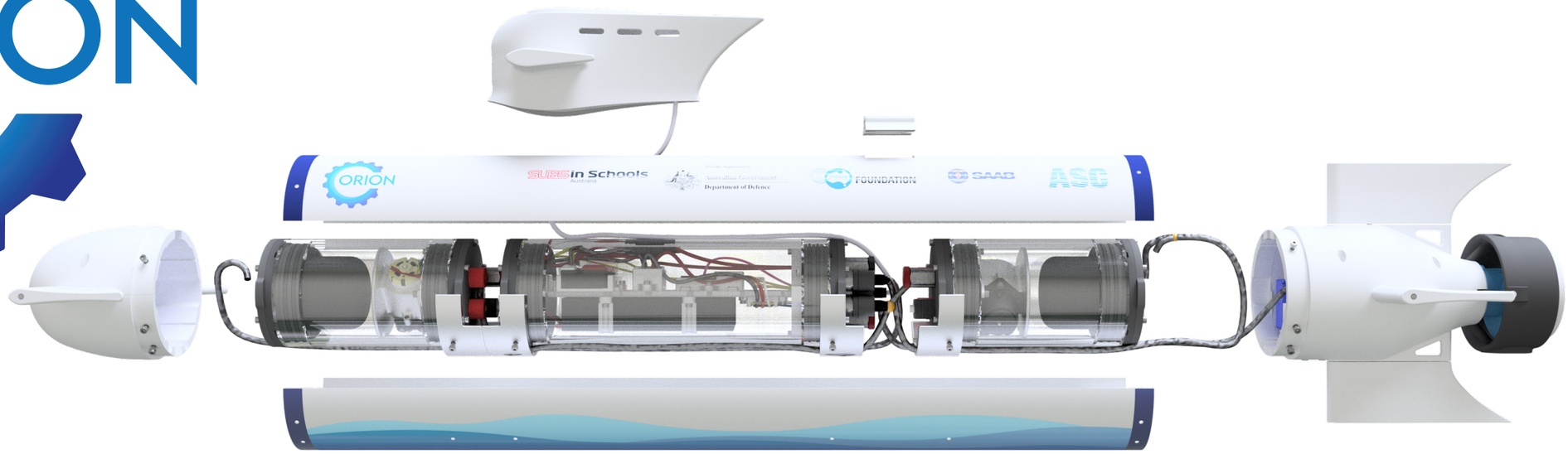
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ORION



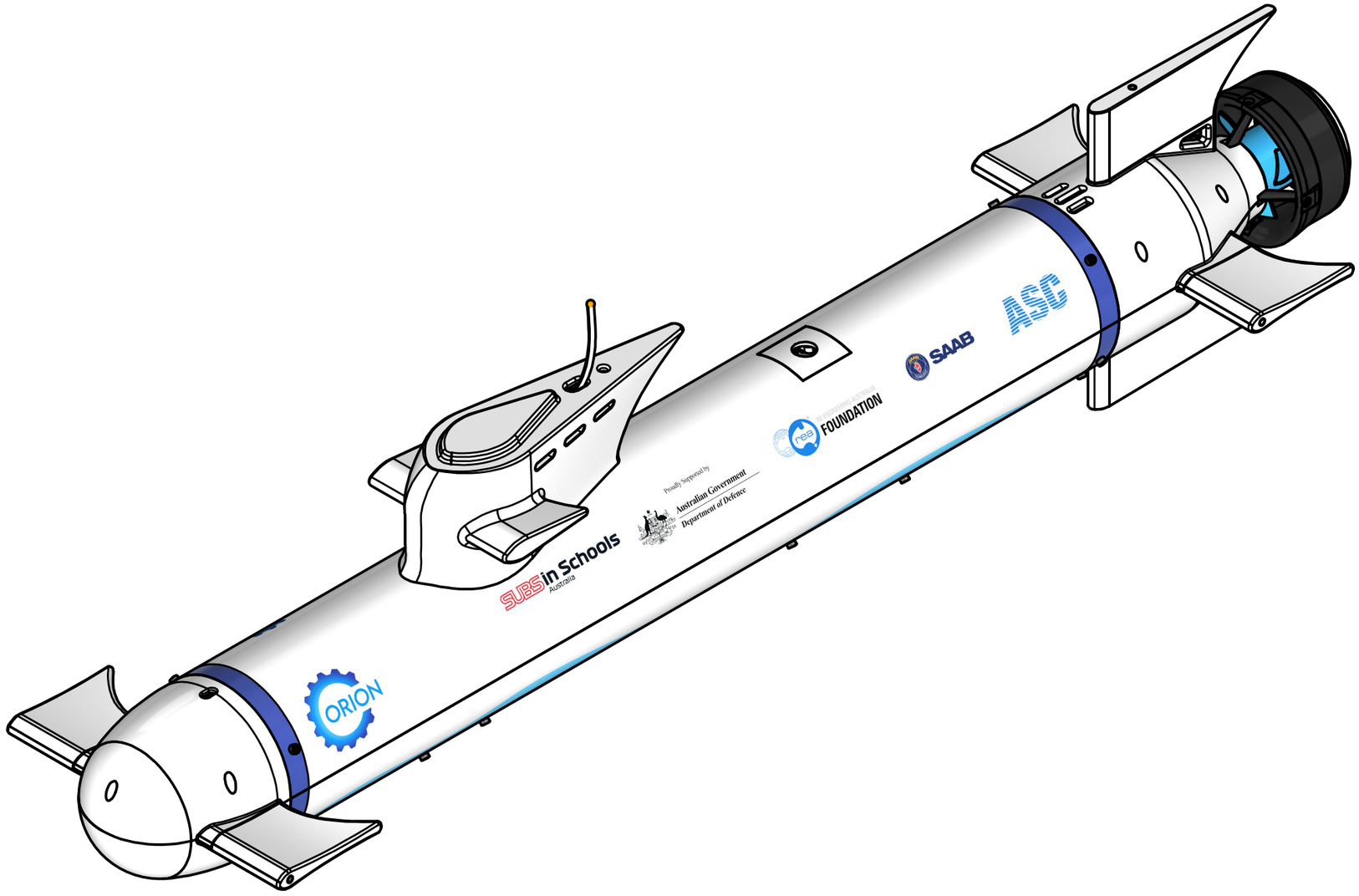
THE HEIGHTS SCHOOL
Preschool - Year 12

SUBS in Schools Australia

COMPLIANCE BOOKLET 2023

Yashwanth Gunasheelan
Deborah Baker
Leon Martin
Vatsal Thakkar

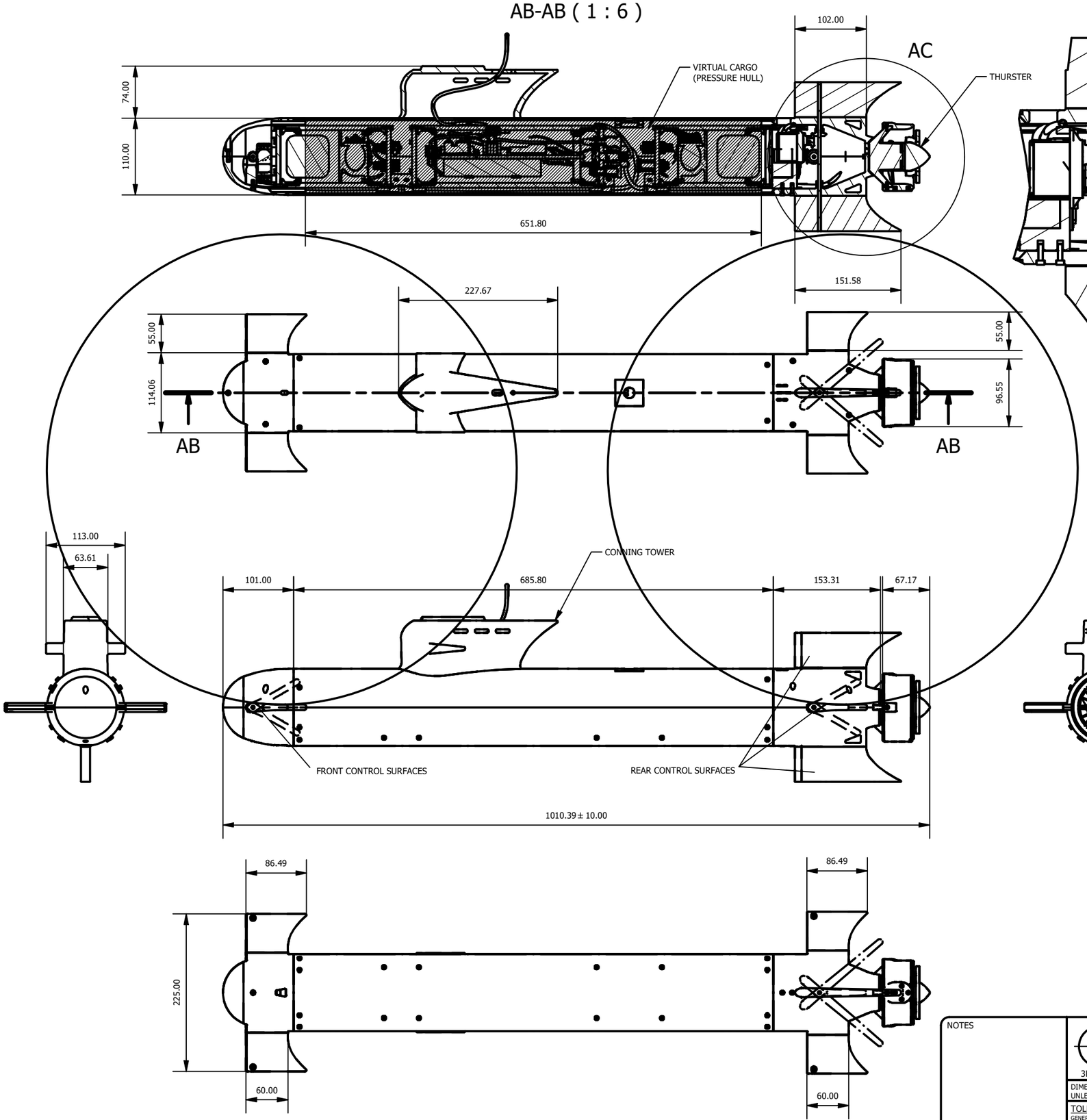
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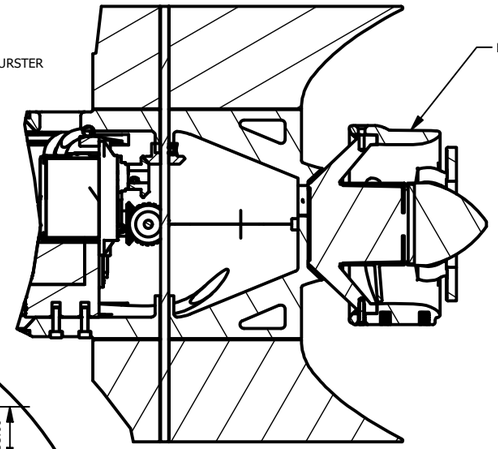
NOTES	3RD ANGLE PROJECTION	DRN ORION	THE HEIGHTS SCHOOL - TEAM ORION
	DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE	DSGN ORION	DOCUMENT NAME: ORION SUBMARINE
	TOLERANCES GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE: ±0.02MM ±0.01°	MFR ORION REV 1 MATERIAL VARIES	SCALE 1 : 3 · PART NAME: Custom Idea Non Rotating V1 · 1/07/2023 SIZE A3 · SHEET 1 OF 27 · REF NO.

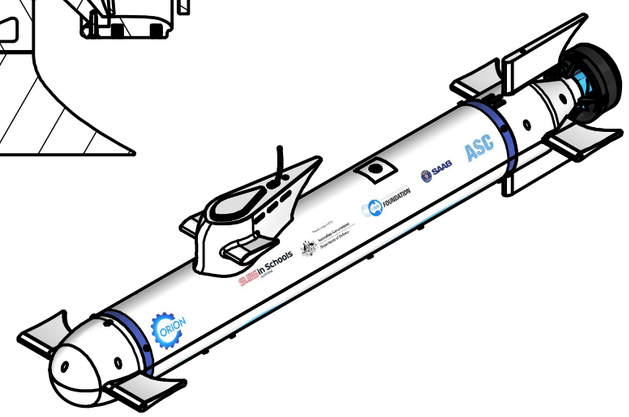
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AB-AB (1 : 6)

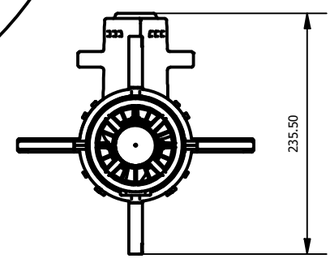


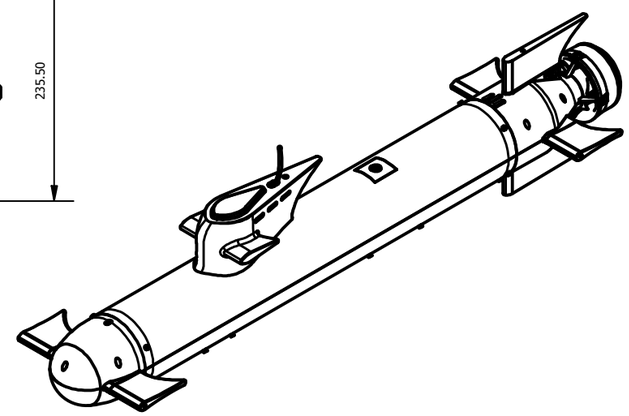
AC (1 : 3)





SHADED ISOMETRIC VIEW (1 : 8)





ISOMETRIC VIEW (1 : 8)

NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS
UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS
UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	ORION
REV	3
MATERIAL	VARIES

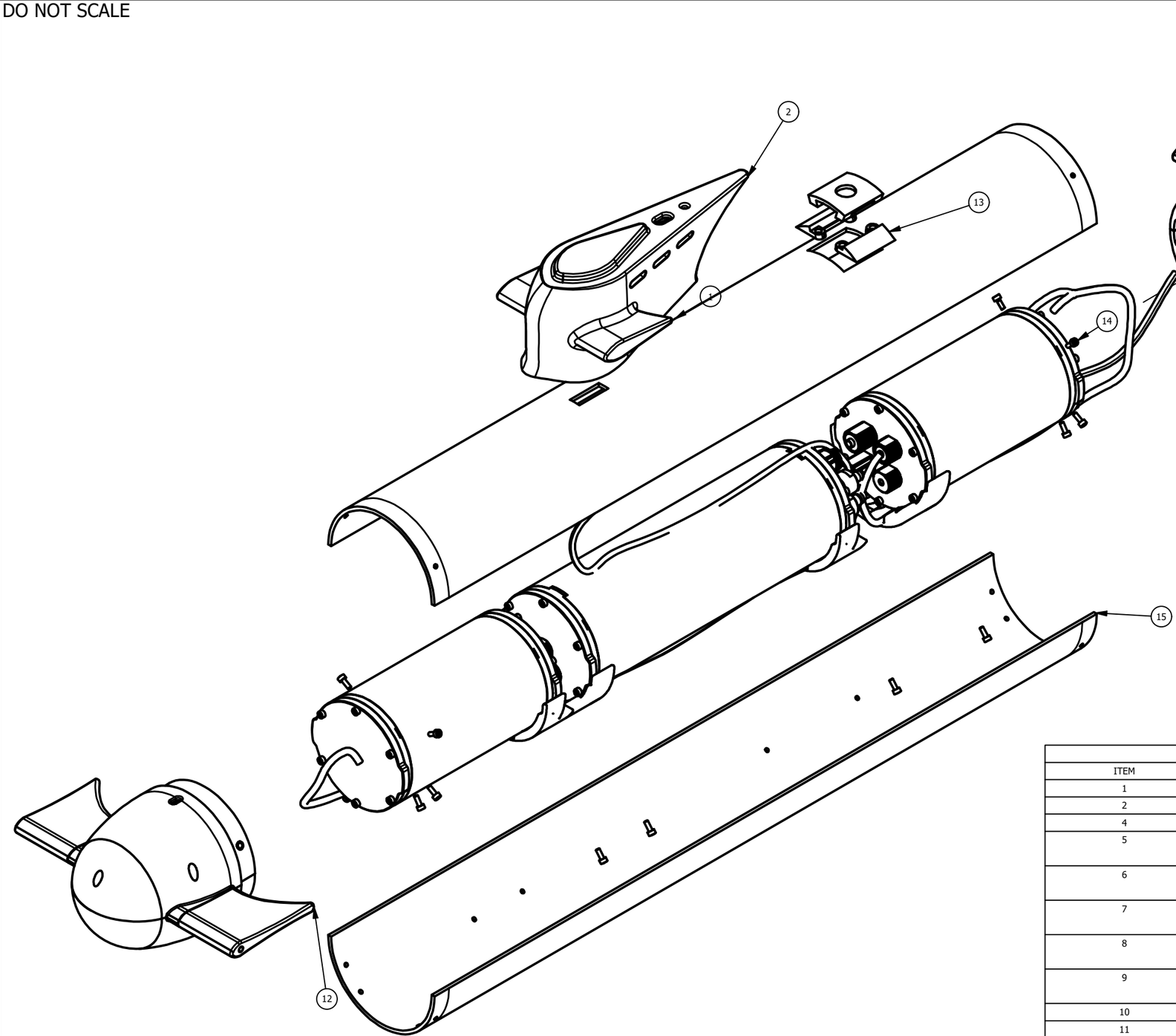
THE HEIGHTS SCHOOL - TEAM ORION

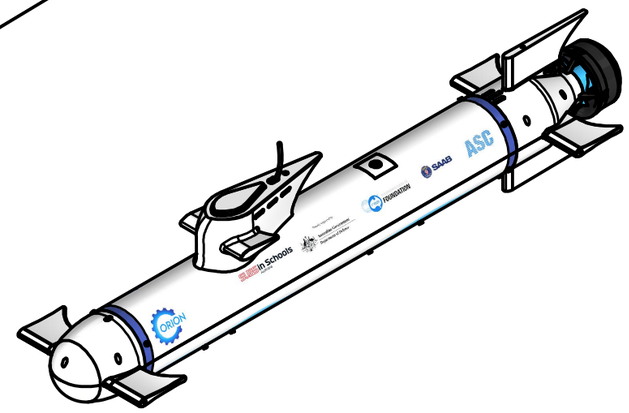
DOCUMENT NAME

SUBMARINE

SCALE 1 : 6	PART NAME Custom Idea Non Rotating V1		1/07/2023
SIZE A3	SHEET 2 OF 27		REF NO.

DO NOT SCALE





SHADED ISOMETRIC (1 : 8)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	Body Pipe Top R1	
2	1	Custom Conning Tower R7	
4	1	PPM Module Assembly R2	
5	4	Hull Suport	DESGINED BY REA. VISIT SITE FOR FULL SPECIFICATIONS.
6	1	COB-123X Cable 3 Pin	DESIGNED MY BLUE TRAIL ENGINEERING. VISIT SITE FOR FULL SPECIFICATIONS.
7	1	COB-124X Cable 4 Pin	DESIGNED MY BLUE TRAIL ENGINEERING. VISIT SITE FOR FULL SPECIFICATIONS.
8	1	COB-126X Cable 6 Pin	DESIGNED MY BLUE TRAIL ENGINEERING. VISIT SITE FOR FULL SPECIFICATIONS.
9	2	COB-128X Cable 8 Pin	DESIGNED MY BLUE TRAIL ENGINEERING. VISIT SITE FOR FULL SPECIFICATIONS.
10	1	Custom Idea Non Rotating V1 Wiring	
11	1	Idea 6 - Non Rotating	
12	1	Custom Fore Cap Assembly V6.ipt	
13	1	Magnetic Latch	
14	20	AS 1420 - 1973 - M3 x 8	ISO METRIC HEXAGON SOCKET HEAD CAP SCREWS
15	1	Body Pipe Bottom R2	
16	2	Ballast Tank Assembly	

NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

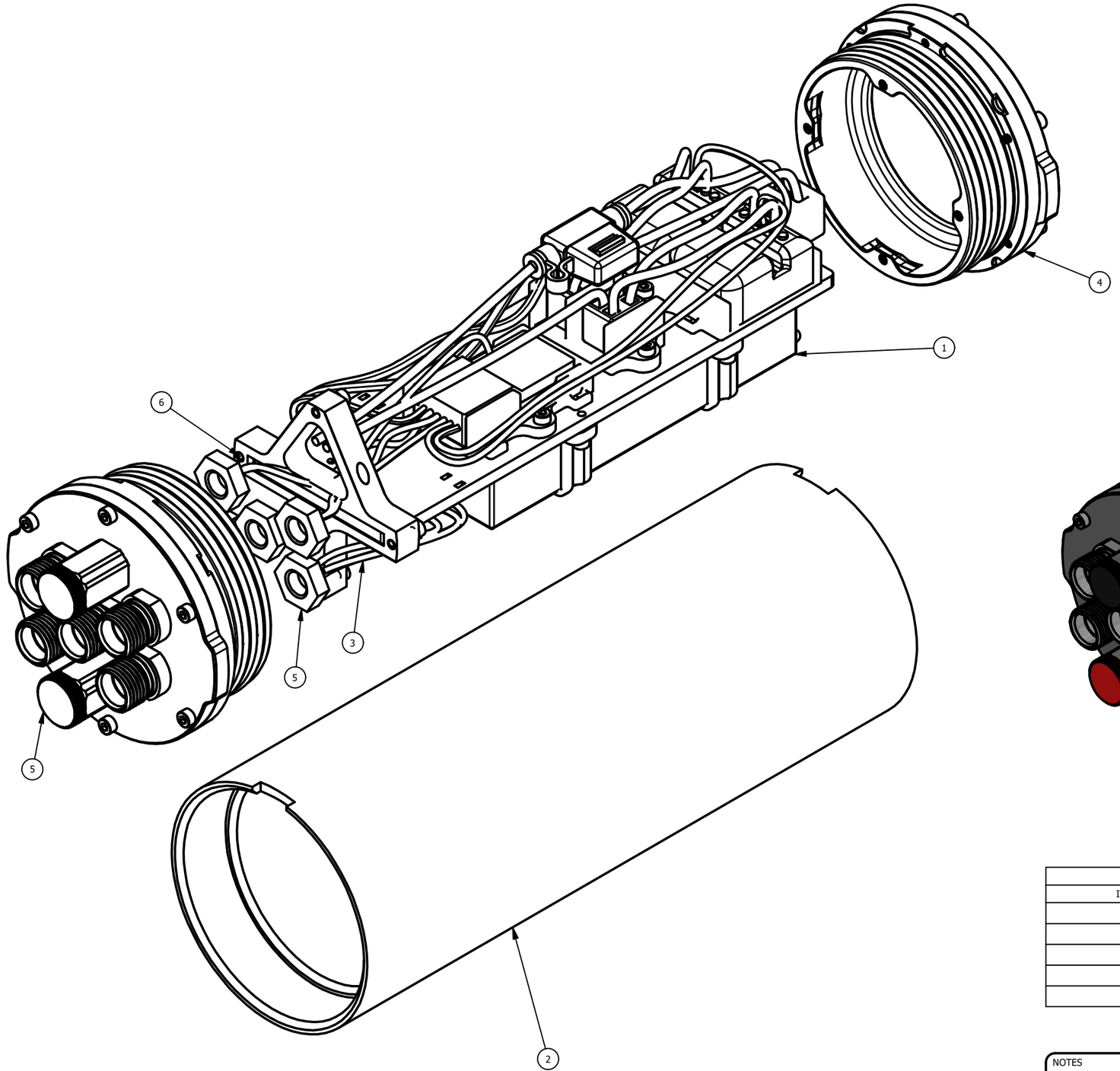
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DSGN	ORION
MFR	ORION
REV	1
MATERIAL	VARIES

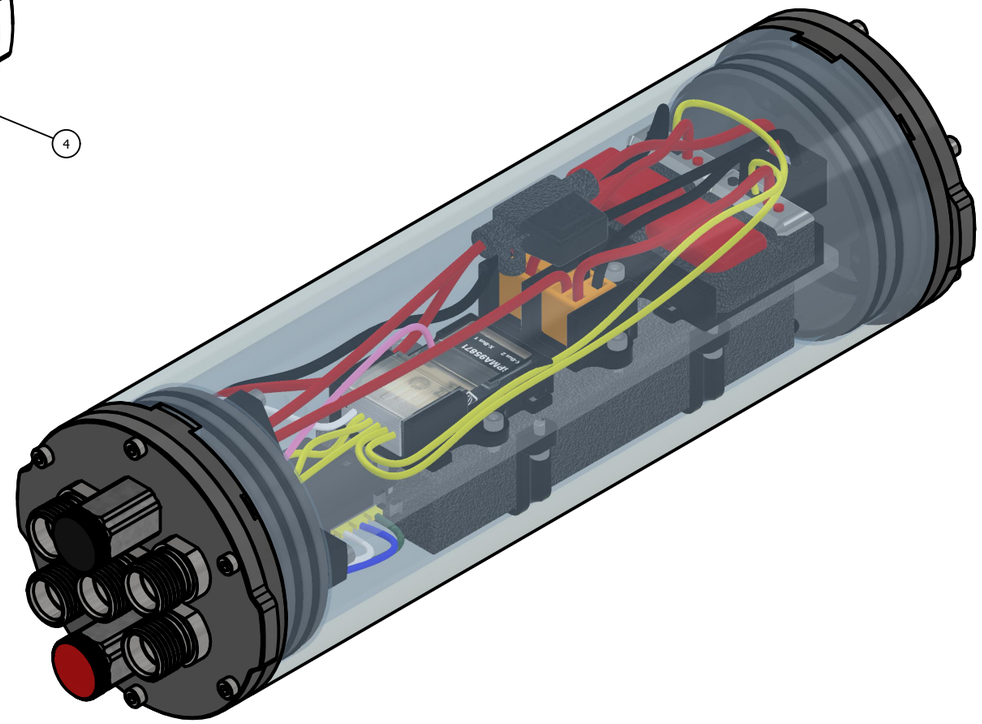
THE HEIGHTS SCHOOL - TEAM ORION

DOCUMENT NAME
SUBMARINE - EXPLODED VIEW

SCALE 3:10	PART NAME Custom Idea Non Rotating V1		1/07/2023
SIZE A3	SHEET 3 OF 27		REF NO.

DO NOT SCALE





SHADED ISOMETRIC (3:5)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	PPM Assembly R2	
2	1	PPM TUBE 250	DESGINED BY BLUE ROBOTICS
4	1	0 Hole Cap Assembly R2	
5	1	7 Hole Cap Assembly R2	
6	4	AS 1420 - 1973 - M3 x 10	ISO METRIC HEXAGON SOCKET HEAD CAP SCREWS

NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

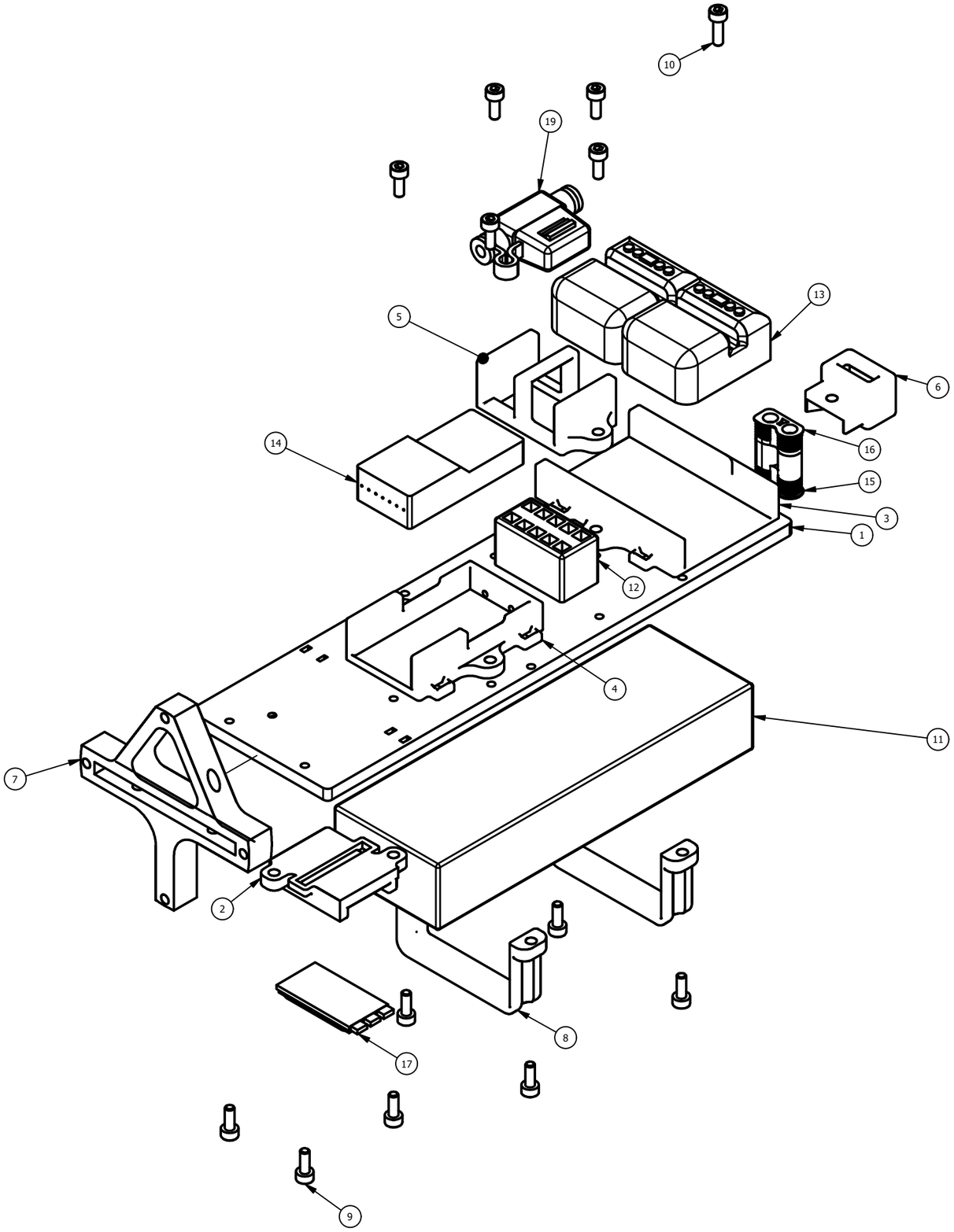
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DSGN	ORION
MFR	MIXED
REV	2
MATERIAL	N/A

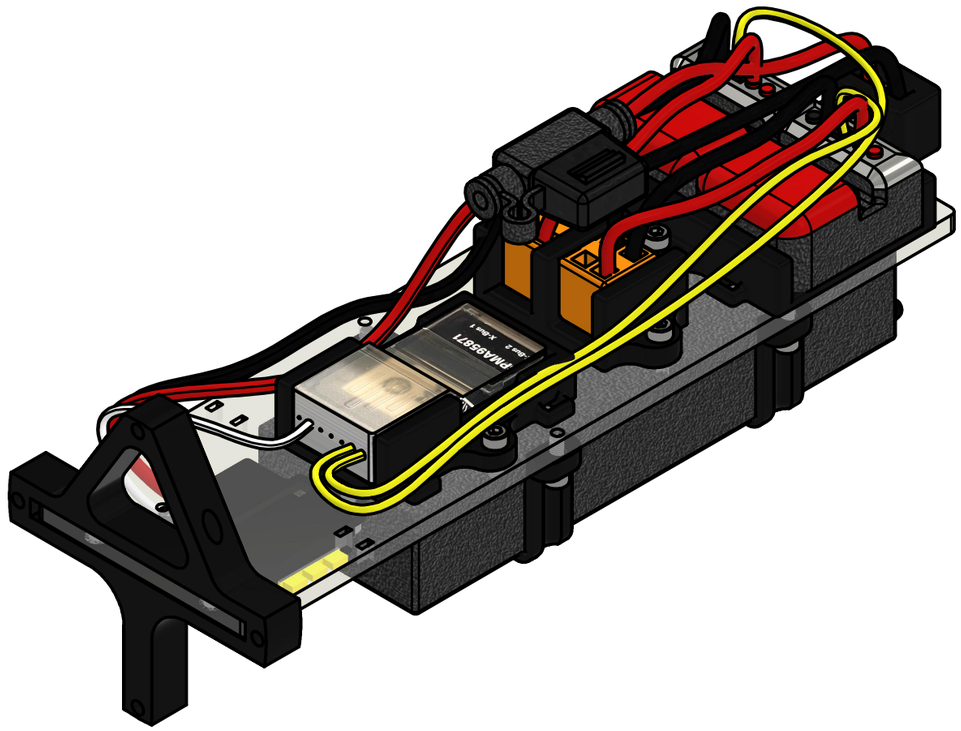
THE HEIGHTS SCHOOL - TEAM ORION

DOCUMENT NAME
PPM MODULE EXPLODED VIEW

SCALE 2:3	PART NAME PPM Module Assembly R2		15/02/2023
SIZE A3	SHEET 4 OF 27		REF NO.

DO NOT SCALE





SHADED ISOMETRIC (4:5)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	ppm plate	
2	1	Thurster ESC Mount	DESIGNED BY REA
3	1	Ballast ESC Mount	DESIGNED BY REA
4	1	Spektrum Reciver Mount	DESIGNED BY REA
5	1	Wago 221 Mount	DESIGNED BY REA
6	1	EC3 Mount	DESIGNED BY REA
7	1	PPM Plate Clamp	DESIGNED BY REA
8	2	Battery Strap	DESIGNED BY REA
9	13	AS 1420 - 1973 - M3 x 8	ISO METRIC HEXAGON SOCKET HEAD CAP SCREWS
10	1	AS 1420 - 1973 - M3 x 10	ISO METRIC HEXAGON SOCKET HEAD CAP SCREWS
11	1	Battery Mockup V1	11.1 VOLT 3300 LITHIUM POLYMER BATTERY
12	1	Wago Mockup V1	WAGO 221 TERMINALS
13	2	Dynamite ESC Mockup V1	DYNAMITE WP BRUSHED ESC
14	1	SPMA95871 GPS Telemetry Sensor	SPEKTRUM AR6610T
15	1	EC3 Connector Socket	EC3 CONNECTOR SOCKET
16	1	EC3 Connector Plug	EC3 CONNECTOR PLUG
17	1	ESC Mockup V1	BLUE ROBOTICS ESC
19	1	Closed Inline Fuse Holder	LITTLEFUSE INLINE FUSE HOLDER

NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	MIXED
REV	1
MATERIAL	MIXED

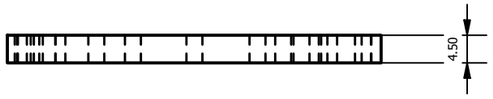
THE HEIGHTS SCHOOL - TEAM ORION

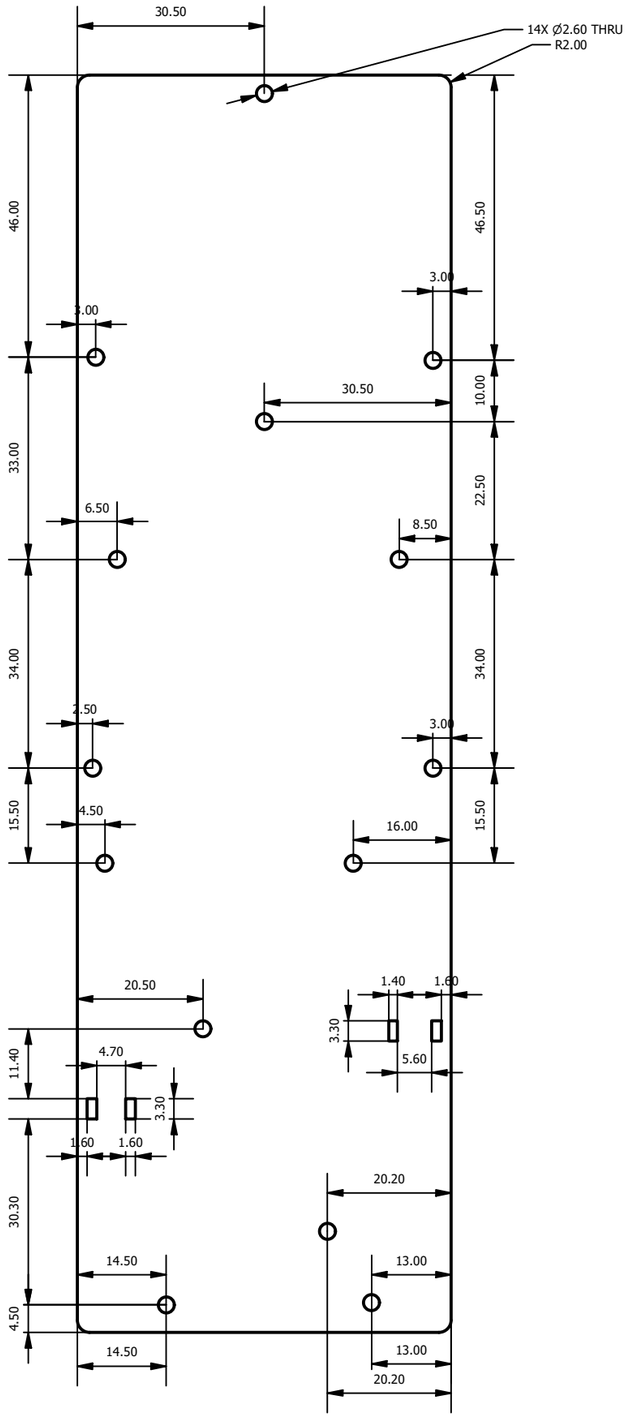
DOCUMENT NAME
PPM ASSEMBLY EXPLODED VIEW

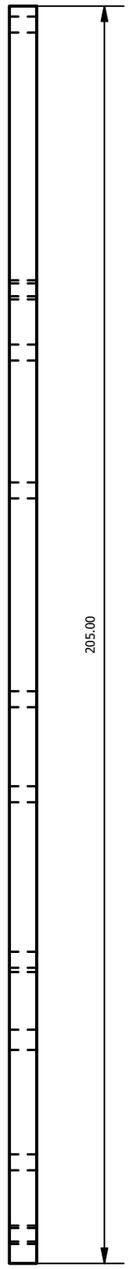
SCALE 2:3 | PART NAME PPM Assembly R2 | 15/02/2023

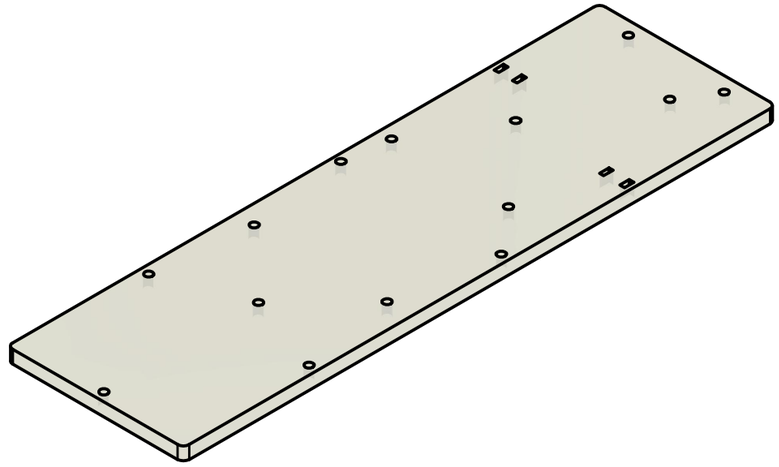
SIZE A3 | SHEET 5 OF 27 | REF NO.

DO NOT SCALE









SHADED ISOMETRIC (2:3)

NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	ORION
REV	2
MATERIAL	ACRYLIC

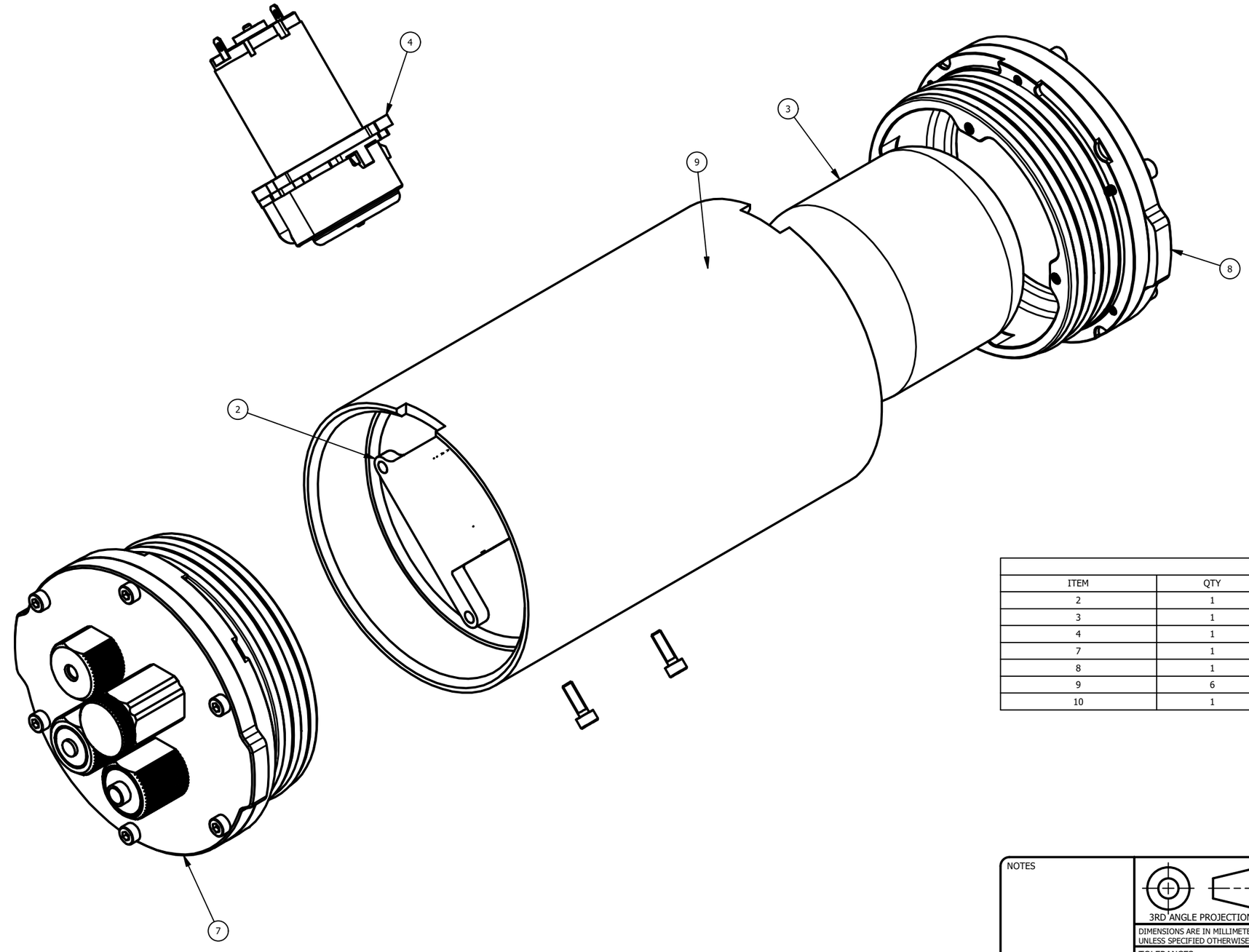
THE HEIGHTS SCHOOL - TEAM ORION

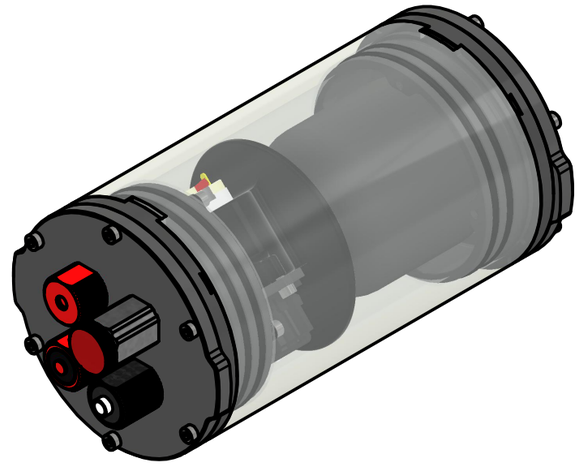
DOCUMENT NAME

PPM PLATE

SCALE 1:1	PART NAME ppm plate		15/06/2021
SIZE A3	SHEET 6 OF 27		REF NO.

DO NOT SCALE





SHADED ISOMETRIC (1:2)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
2	1	Ballast Tank Pump Mount	DESGINED BY REA
3	1	Bladder	SILICON RUBBER BLOOD BLADDER
4	1	Final Peristalic Pump Assembly	DESIGNED AND MANUFACTURED BY INTLLAB
7	1	4 Hole Cap Assembly R2	
8	1	0 Hole End Cap Assembly	
9	6	AS 1420 - 1973 - M3 x 10	ISO METRIC HEXAGON SOCKET HEAD CAP SCREWS
10	1	BALLAST TANK TUBE	

NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

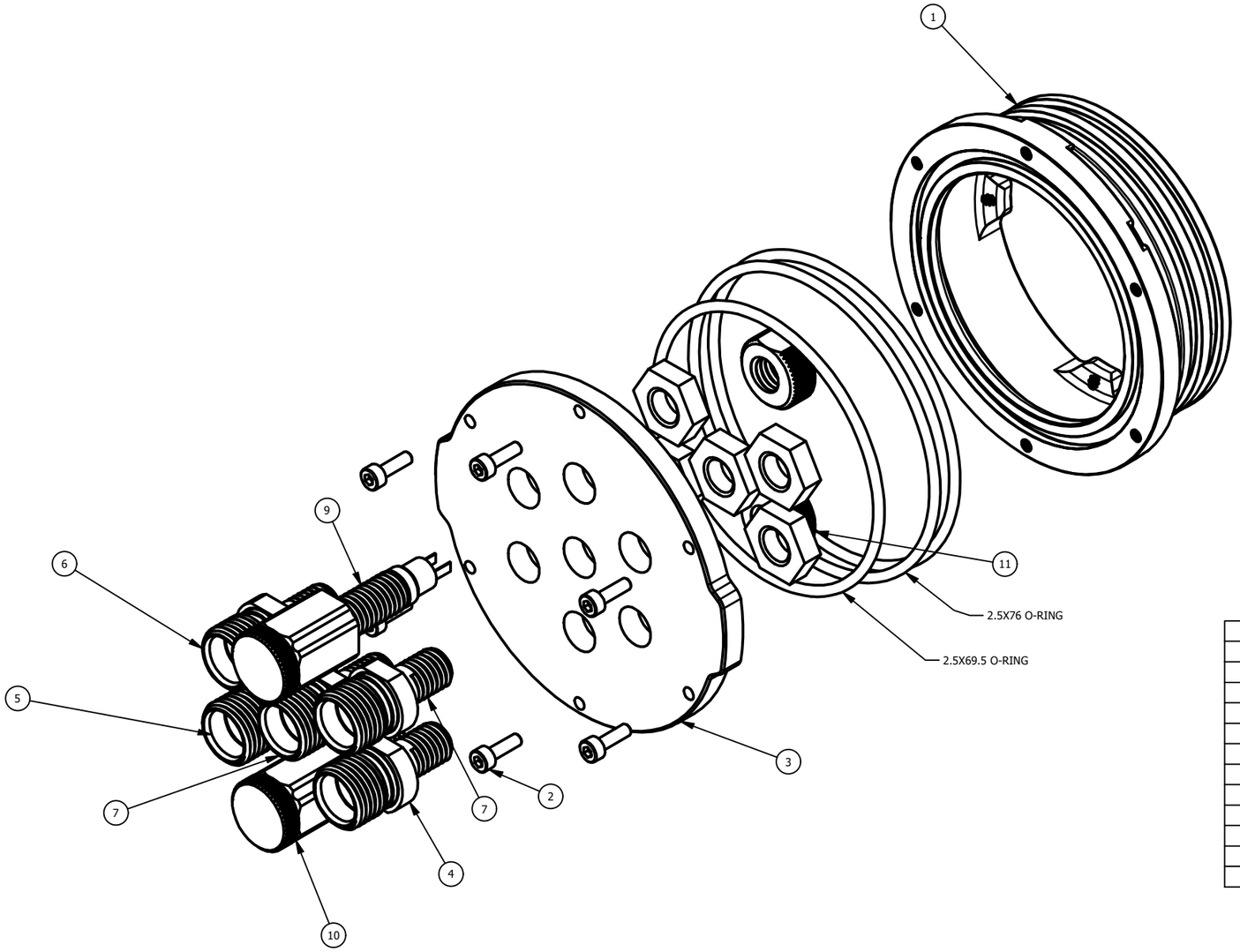
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DSGN	ORION
MFR	MIXED
REV	2
MATERIAL	MIXED

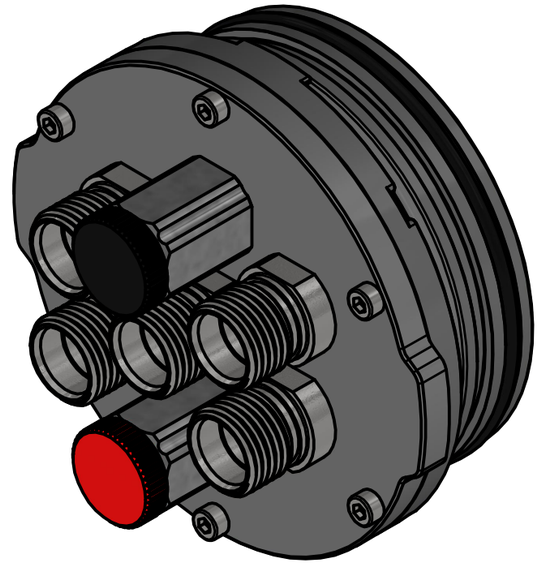
THE HEIGHTS SCHOOL - TEAM ORION

DOCUMENT NAME
BALLAST TANK EXPLODED VIEW

SCALE 8:10	PART NAME Ballast Tank Assembly		15/02/2023
SIZE A3	SHEET 7 OF 27		REF NO.

DO NOT SCALE





SHADED ISOMETRIC (1:1)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	WTE3-M-LOCKING-FLANGE-SEAL-R1-RP	DESGINED BY BLUE ROBOTICS
2	6	AS 1420 - 1973 - M3 x 10	ISO METRIC HEXAGON SOCKET HEAD SCREWS
3	1	WTE3-M-END-CAP-7-HOLE-R1	DESGINED BY BLUE ROBOTICS
4	1	COB-1130 Bulkhead 3 Pin	DESGINED BY BLUE ROBOTICS
5	1	COB-1140 Bulkhead 4 Pin	DESGINED BY BLUE ROBOTICS
6	1	COB-1160 Bulkhead 6 Pin	DESGINED BY BLUE ROBOTICS
7	2	COB-1180 Bulkhead 8 Pin	DESGINED BY BLUE ROBOTICS
9	1	SWITCH-M10-5A-R1-RP	DESGINED BY BLUE ROBOTICS
10	1	VENT-M10-R1-RP	DESGINED BY BLUE ROBOTICS
11	2	PENETRATOR-M-NUT-10-8-R2	DESGINED BY BLUE ROBOTICS
12	5	AS 1474 - M10	Hex Nut

NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS
UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS
UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

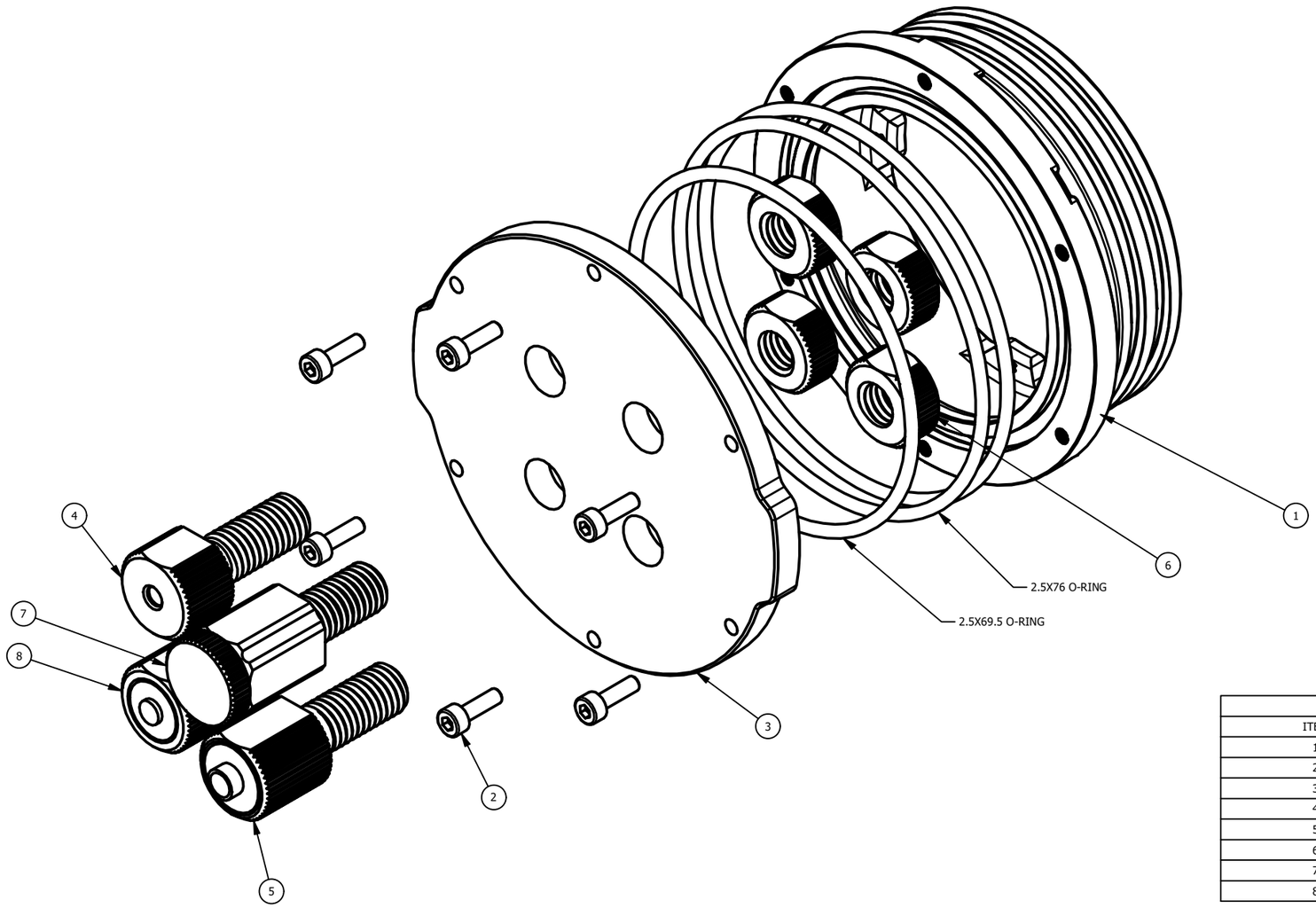
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DSGN	ORION
MFR	MIXED
REV	2
MATERIAL	MIXED

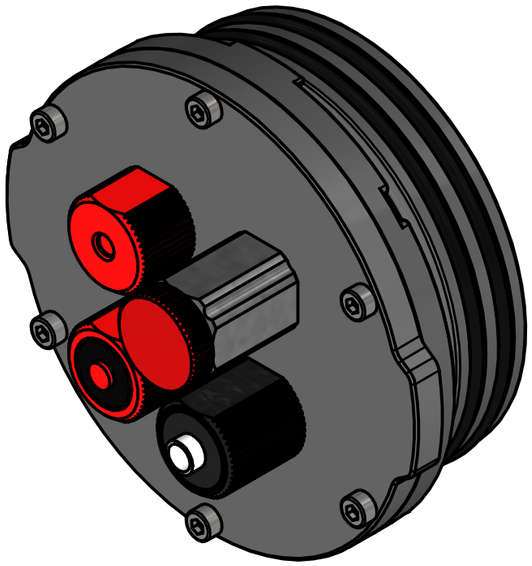
THE HEIGHTS SCHOOL - TEAM ORION

DOCUMENT NAME
SEVEN HOLE END CAP EXPLODED VIEW

SCALE 8:10	PART NAME 7 Hole Cap Assembly R2		19/05/2023
SIZE A3	SHEET 8 OF 27		REF NO.

DO NOT SCALE





SHADED ISOMETRIC (1 : 1)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	WTE3-M-LOCKING-FLANGE-SEAL-R1-RP	DESGINED BY BLUE ROBOTICS
2	6	AS 1420 - 1973 - M3 x 10	ISO METRIC HEXAGON SOCKET HEAD SCREWS
3	1	WTE3-M-END-CAP-4-HOLE-R1	DESGINED BY BLUE ROBOTICS
4	1	PENETRATOR-M-BLANK-10-25-R3	DESGINED BY BLUE ROBOTICS
5	1	PENETRATOR-M-BOLT-8MM-10-25-R1	DESGINED BY BLUE ROBOTICS
6	4	PENETRATOR-M-NUT-10-8-R2	DESGINED BY BLUE ROBOTICS
7	1	VENT-M10-R1-RP	DESGINED BY BLUE ROBOTICS
8	1	PENETRATOR-M-BOLT-6MM-10-25-R2	DESGINED BY BLUE ROBOTICS

NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS
UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS
UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

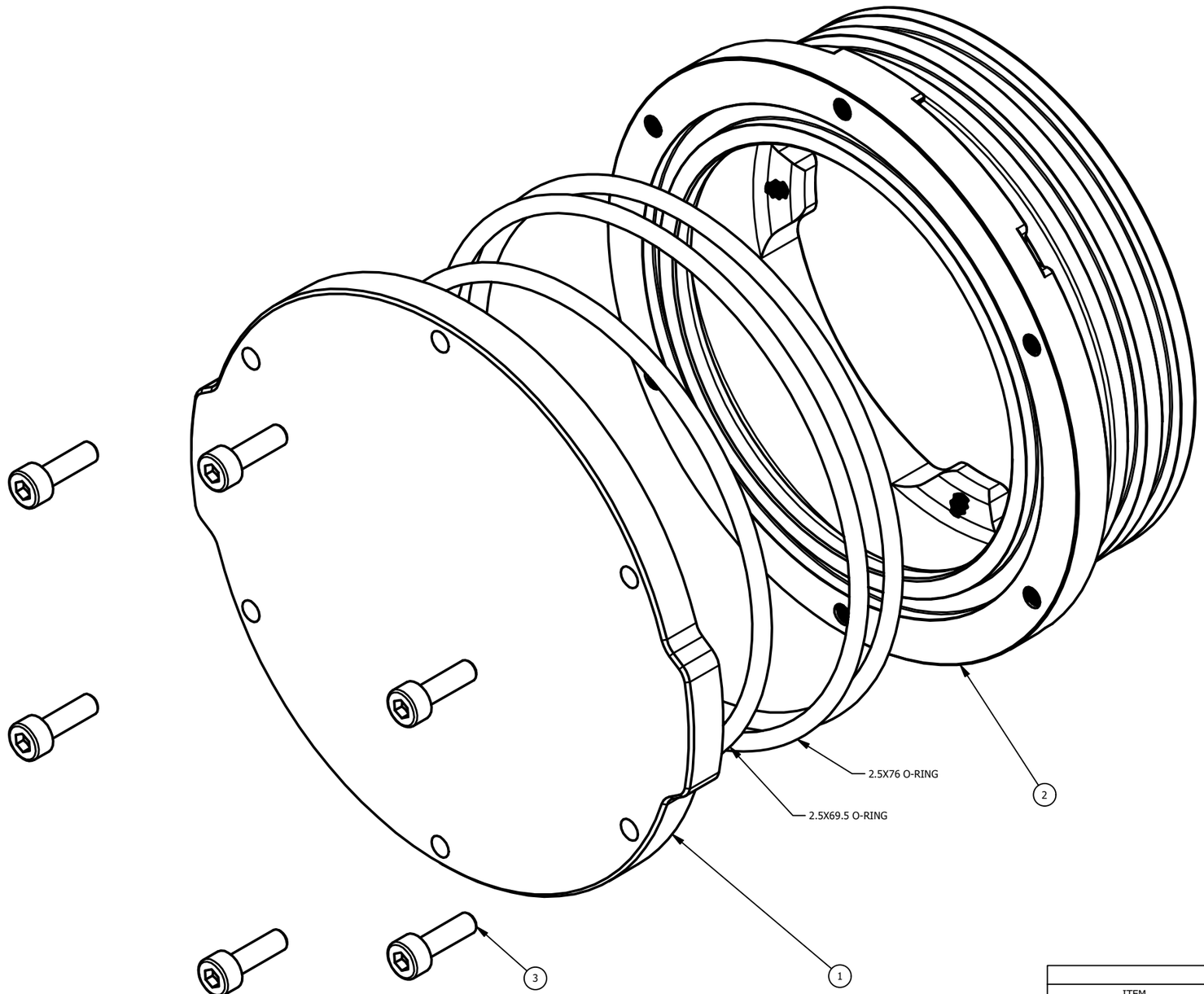
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DSGN	ORION
MFR	MIXED
REV	2
MATERIAL	MIXED

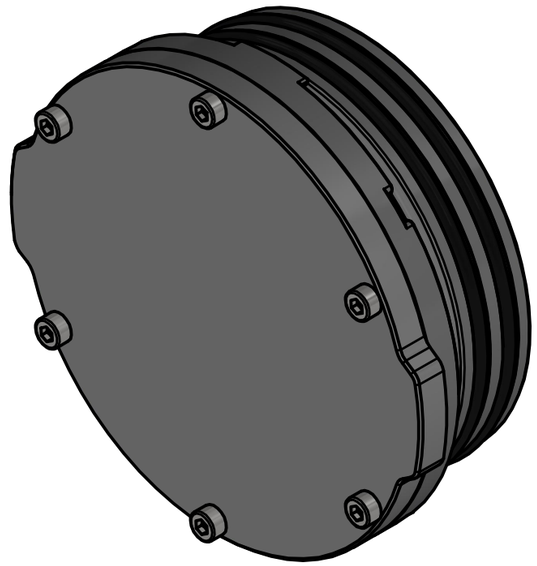
THE HEIGHTS SCHOOL - TEAM ORION

DOCUMENT NAME
FOUR HOLE END CAP EXPLODED VIEW

SCALE 1:1	PART NAME 4 Hole Cap Assembly R2		19/05/2023
SIZE A3	SHEET 9 OF 27		REF NO.

DO NOT SCALE





SHADED ISOMETRIC (1:1)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	WTE3-M-END-CAP-R1	DESIGNED BY BLUE ROBOTICS
2	1	WTE3-M-LOCKING-FLANGE-SEAL-R1-RP	DESIGNED BY BLUE ROBOTICS
3	6	AS 1420 - 1973 - M3 x 10	ISO metric hexagon socket head cap screws

NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

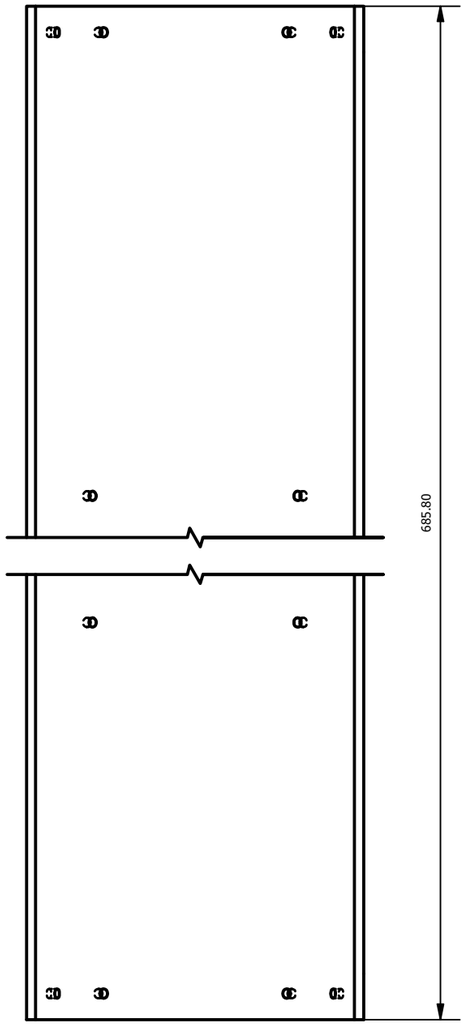
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DSGN	ORION
MFR	MIXED
REV	2
MATERIAL	MIXED

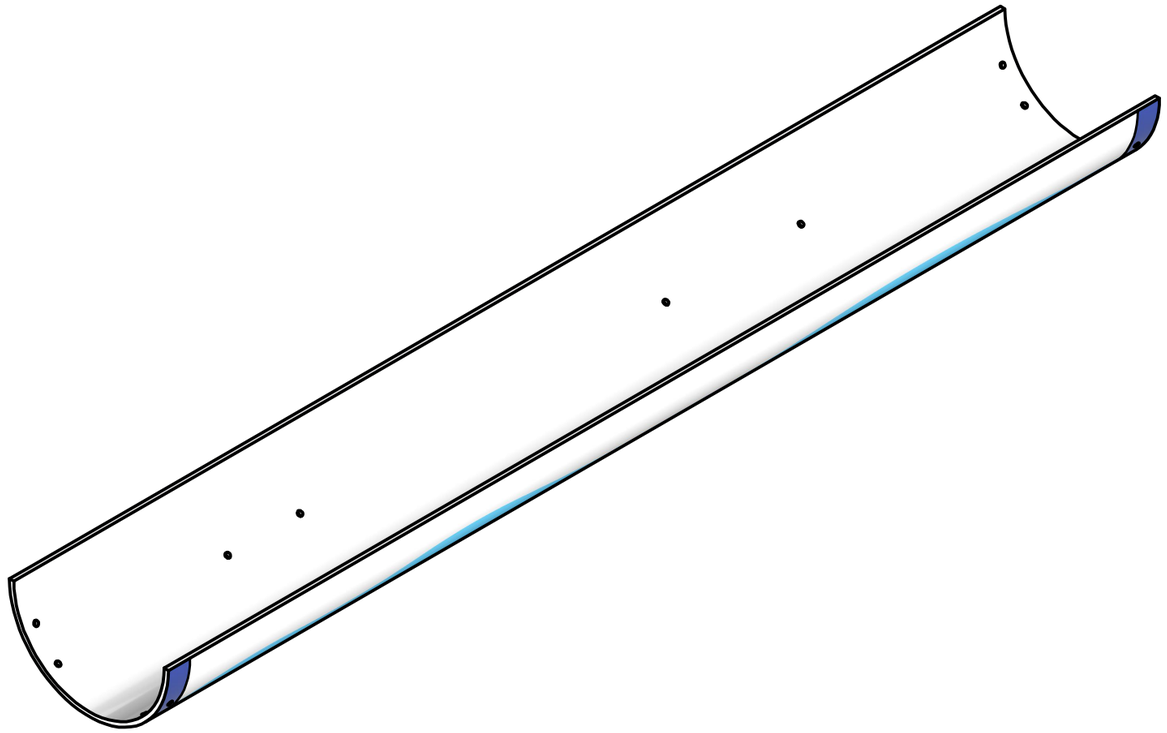
THE HEIGHTS SCHOOL - TEAM ORION

DOCUMENT NAME
ZERO END CAP EXPLODED VIEW

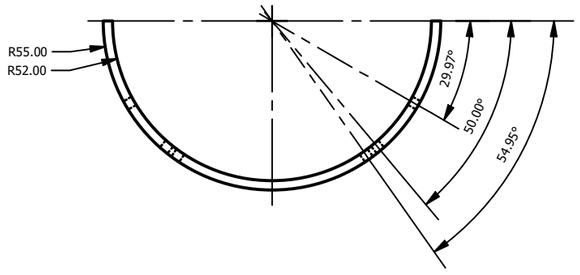
SCALE 1:1	PART NAME 0 Hole Cap Assembly R2		19/05/2023
SIZE A3	SHEET 10 OF 27		REF NO.

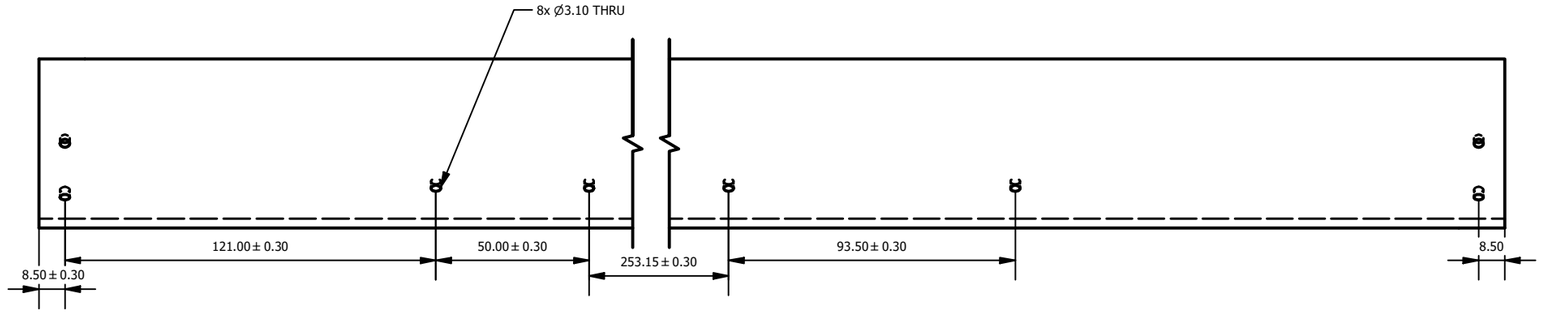
DO NOT SCALE





SHADED ISOMETRIC (1 : 3)





NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	VARIOUS
REV	2
MATERIAL	PVC

ORION | SUBS in Schools Australia | THE HEIGHTS SCHOOL Preschool - Year 12

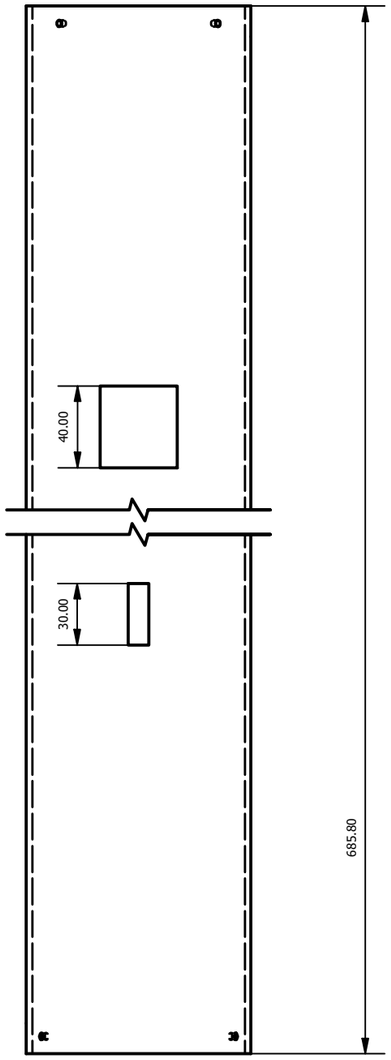
THE HEIGHTS SCHOOL - TEAM ORION

DOCUMENT NAME

BODY HULL BOTTOM

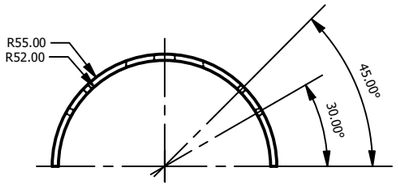
SCALE 1 : 2	PART NAME Body Pipe Bottom R2		11/08/2023
SIZE A3	SHEET 11 OF 27		REF NO.

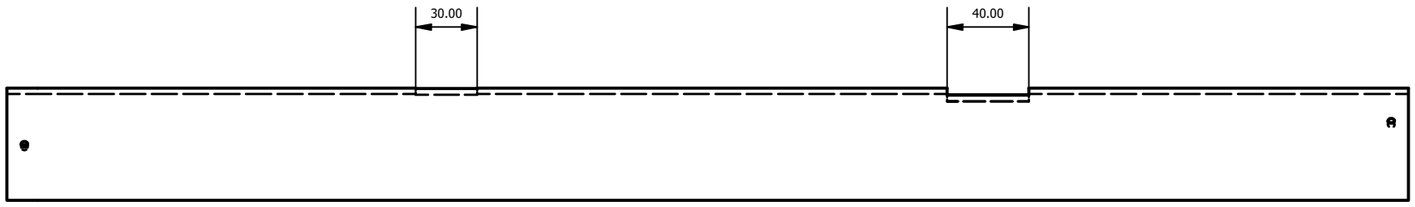
DO NOT SCALE





SHADED ISOMETRIC (1 : 3)





NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	VARIOUS
REV	1
MATERIAL	PVC

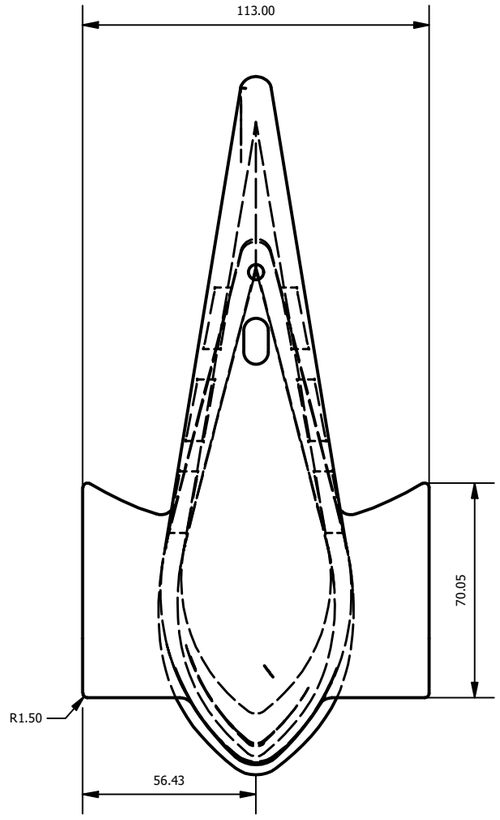
THE HEIGHTS SCHOOL - TEAM ORION

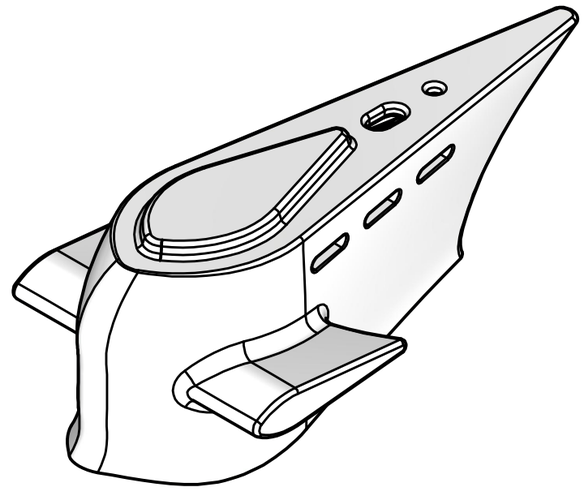
DOCUMENT NAME

BODY HULL TOP

SCALE 1:3	PART NAME Body Pipe Top R1		16/06/2023
SIZE A3	SHEET 12 OF 27		REF NO.

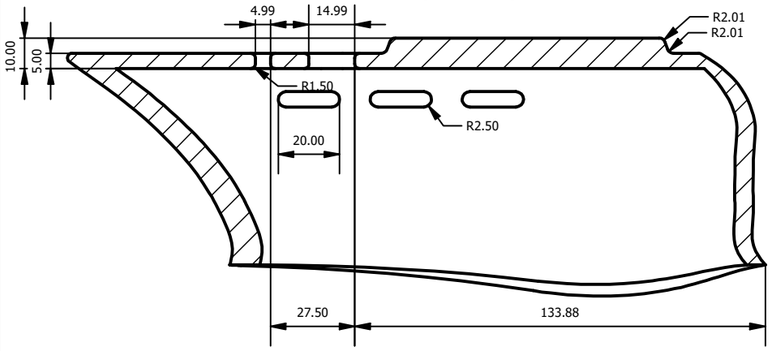
DO NOT SCALE

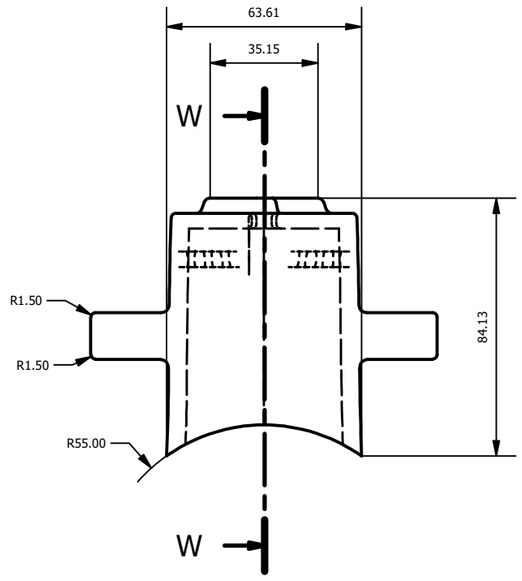


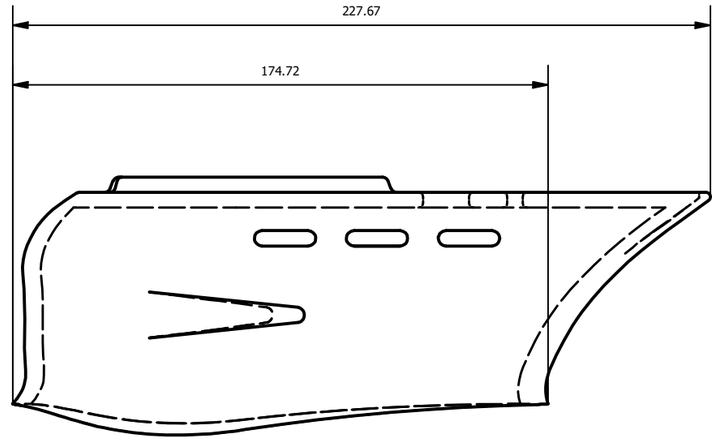


SHADED ISOMETRIC (1 : 2)

W-W (1 : 2)







NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	ORION
REV	7
MATERIAL	ABS

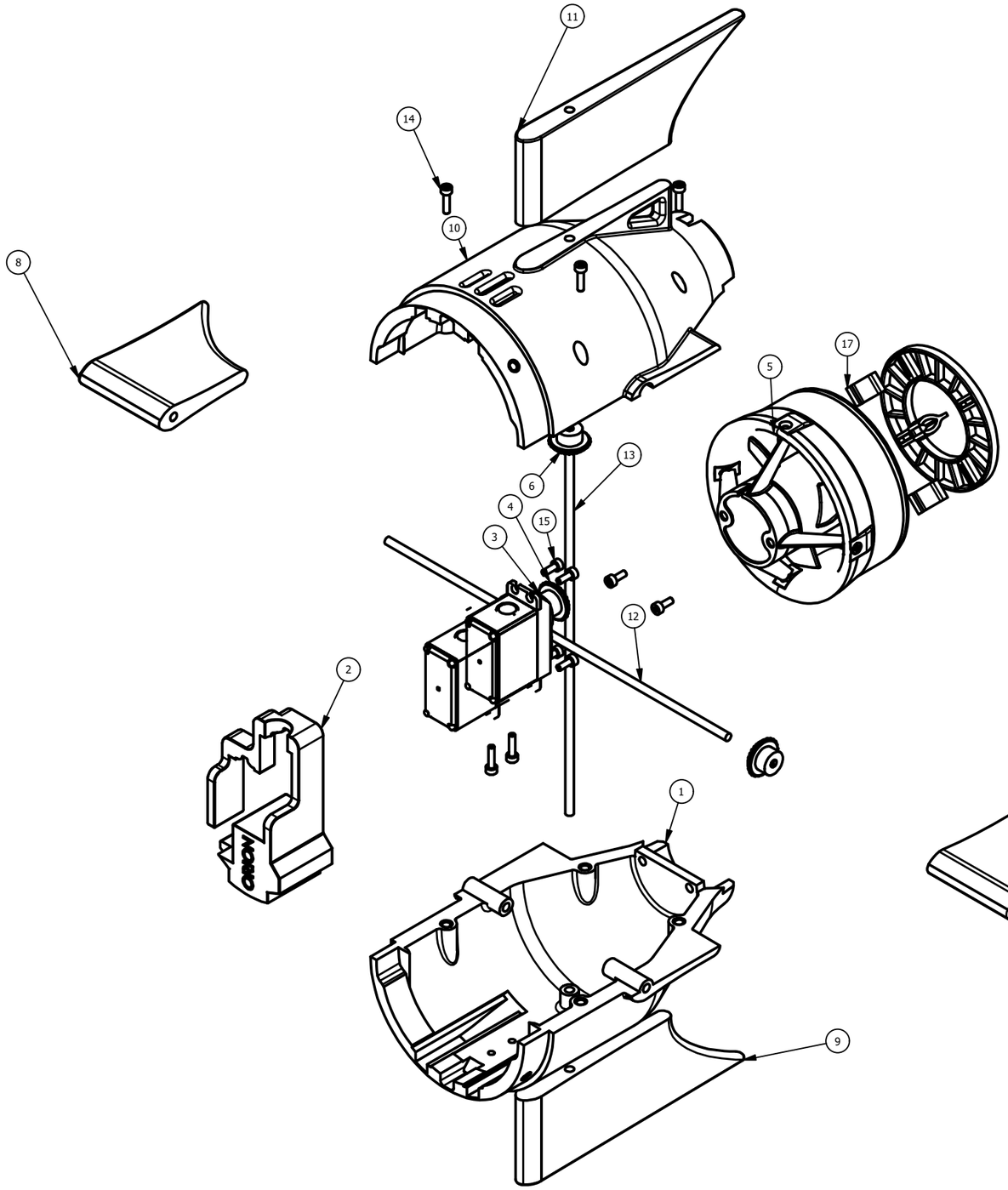
THE HEIGHTS SCHOOL - TEAM ORION

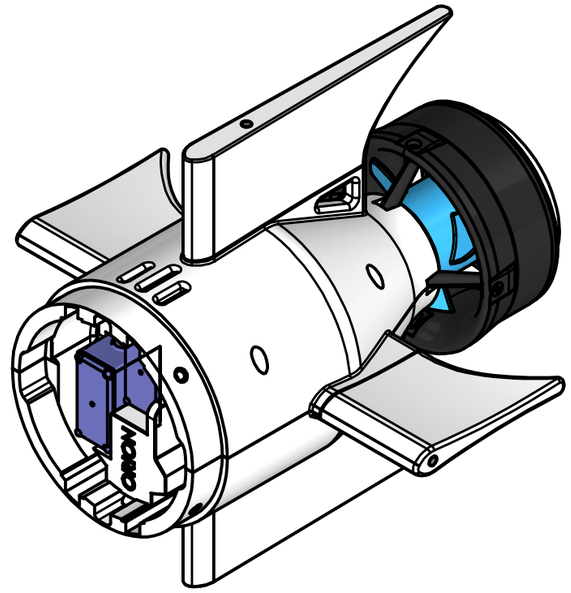
DOCUMENT NAME

CONNING TOWER

SCALE 1 : 2	PART NAME Custom Conning Tower R7		1/07/2023
SIZE A3	SHEET 13 OF 27		REF NO.

DO NOT SCALE





SHADED ISOMETRIC (4:10)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	Custom Aft Cap V6	
2	1	Custom Servo Holder V3	
3	2	Servo Assembly	
4	2	Bevel Gear 6mm Bore	STEP AP203
5	1	T200 Thruster Body	STEP AP214
6	2	Bevel Gear 4mm Bore	STEP AP203
7	1	Custom Short Control Plane V6	
8	1	Custom Short Control Plane Starboard V6	
9	1	Custom Long Control Plane Bottom V6	
10	1	Custom Aft Cap Top V6	
11	1	Custom Long Control Plane Top V6	
12	1	Custom Aft Cap Rod Horizontal V6	
13	1	Custom Aft Cap Rod Vertical V6	
14	6	AS 1420 - 1973 - M3 x 12	ISO metric hexagon socket head cap screws
15	8	AS 1420 - 1973 - M3 x 8	ISO metric hexagon socket head cap screws
16	4	AS 1421 - M3 x 4 Cup Point	ISO metric hexagon socket set screws
17	1	Shroud Rear With Clips	

NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	ORION
REV	6
MATERIAL	VARIES

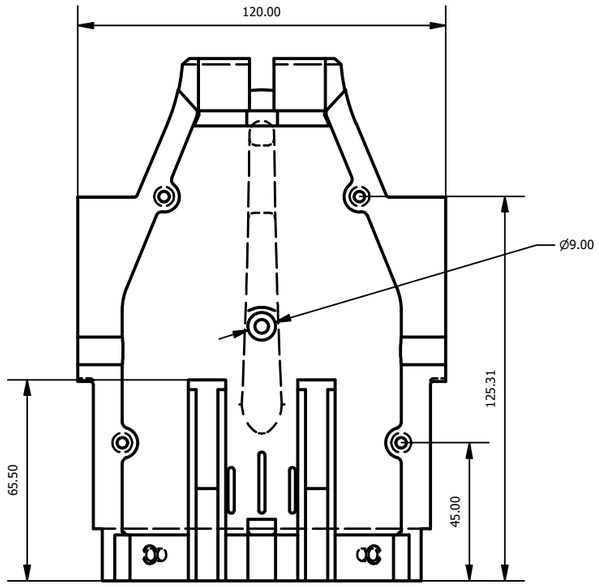
THE HEIGHTS SCHOOL - TEAM ORION

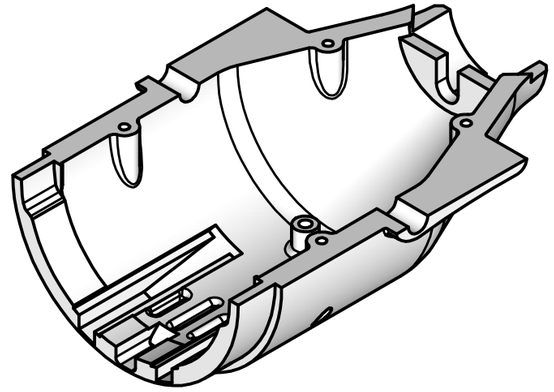
DOCUMENT NAME

AFT CAP

SCALE 4:10	PART NAME Idea 6 - Non Rotating		29/06/2023
SIZE A3	SHEET 14 OF 27		REF NO.

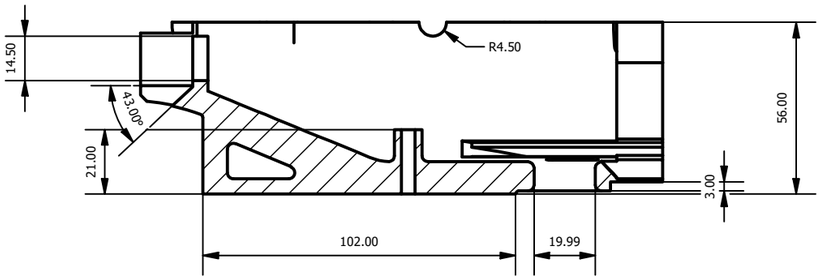
DO NOT SCALE

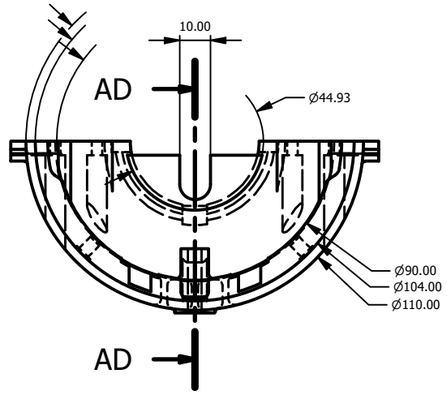


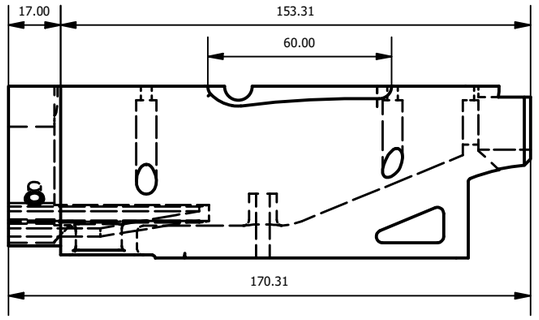


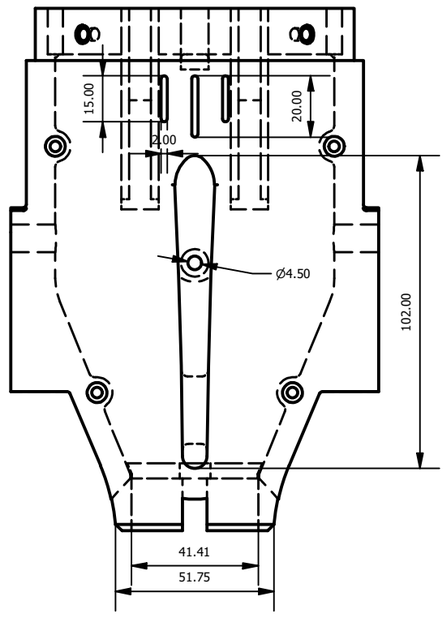
SAHDED ISOMETRIC (1 : 2)

AD-AD (1 : 2)









NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	ORION
REV	6
MATERIAL	ASA PLASTIC

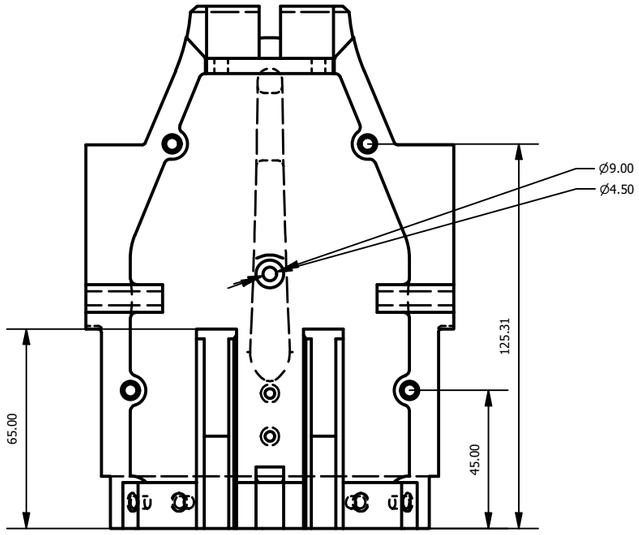
THE HEIGHTS SCHOOL - TEAM ORION

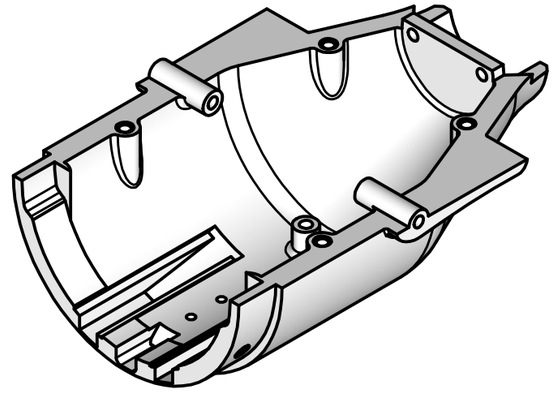
DOCUMENT NAME

AFT CAP TOP

SCALE 1 : 2	PART NAME Custom Aft Cap Top V6		14/04/2023
SIZE A3	SHEET 15 OF 27		REF NO.

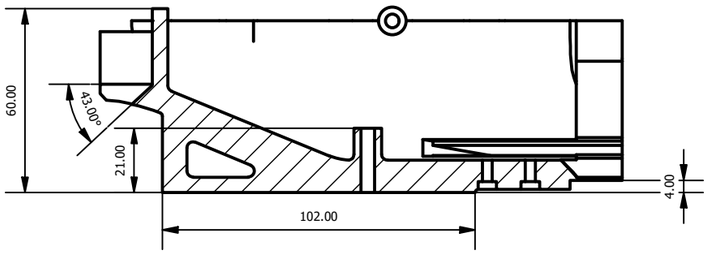
DO NOT SCALE

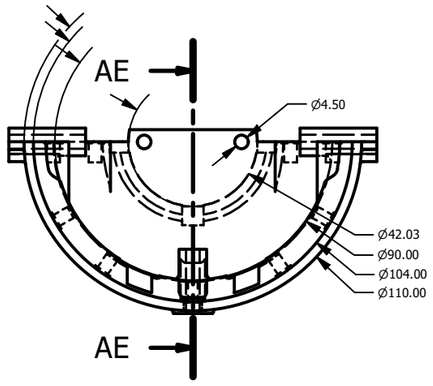


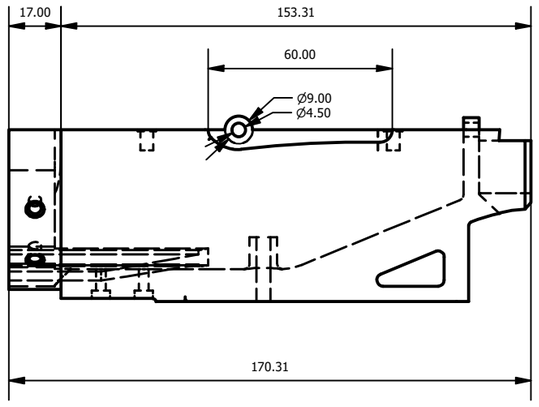


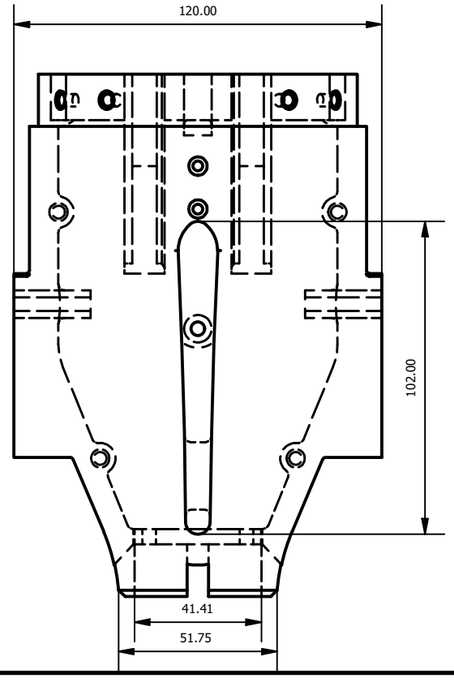
AFT CAP BOTTOM (1 : 2)

AE-AE (1 : 2)



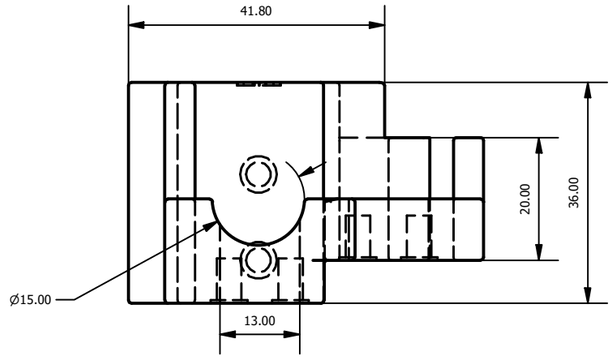


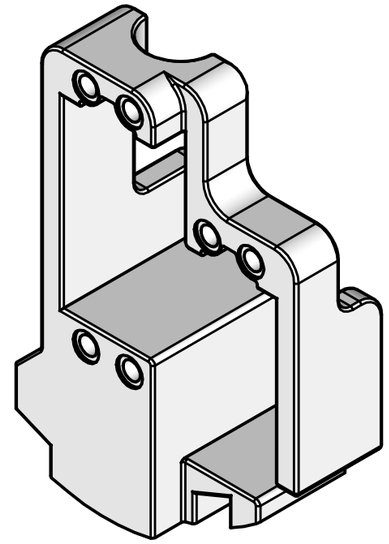




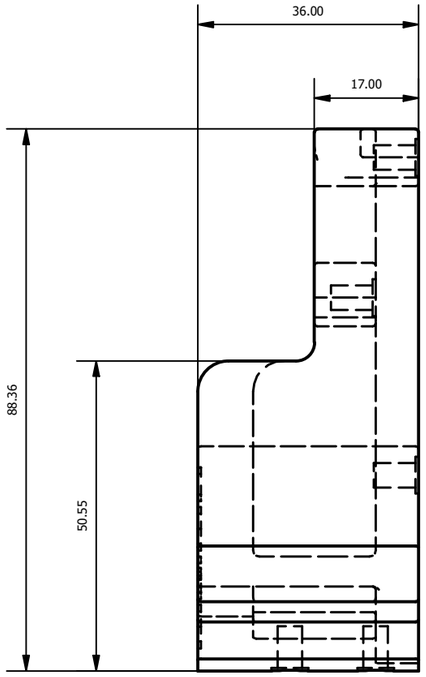
NOTES		DRN ORION	THE HEIGHTS SCHOOL - TEAM ORION
	3RD ANGLE PROJECTION	DSGN ORION	DOCUMENT NAME: AFT CAP BOTTOM
	DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE	MFR ORION	
	TOLERANCES: GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE: ±0.02MM ±0.01°	REV 6	
		MATERIAL ASA PLASTIC	
		SCALE 1 : 2	PART NAME: Custom Aft Cap V6
		SIZE A3	SHEET 16 OF 27
			19/03/2023
			REF NO.

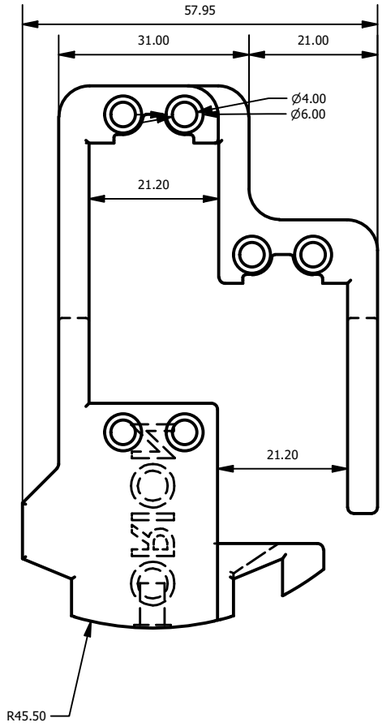
DO NOT SCALE

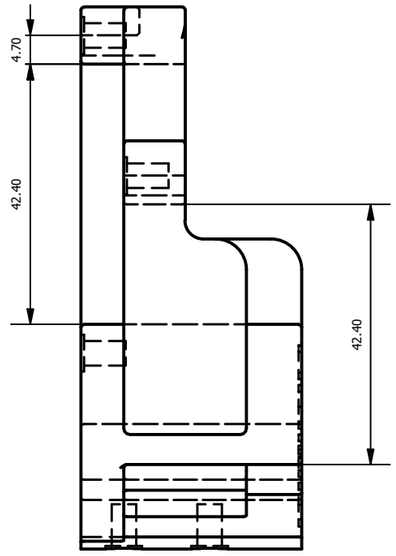


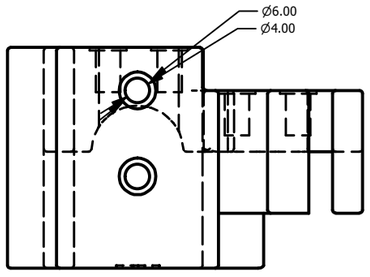


SHADED ISOMETRIC (1 : 1)









NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	ORION
REV	2
MATERIAL	ASA PLASTIC

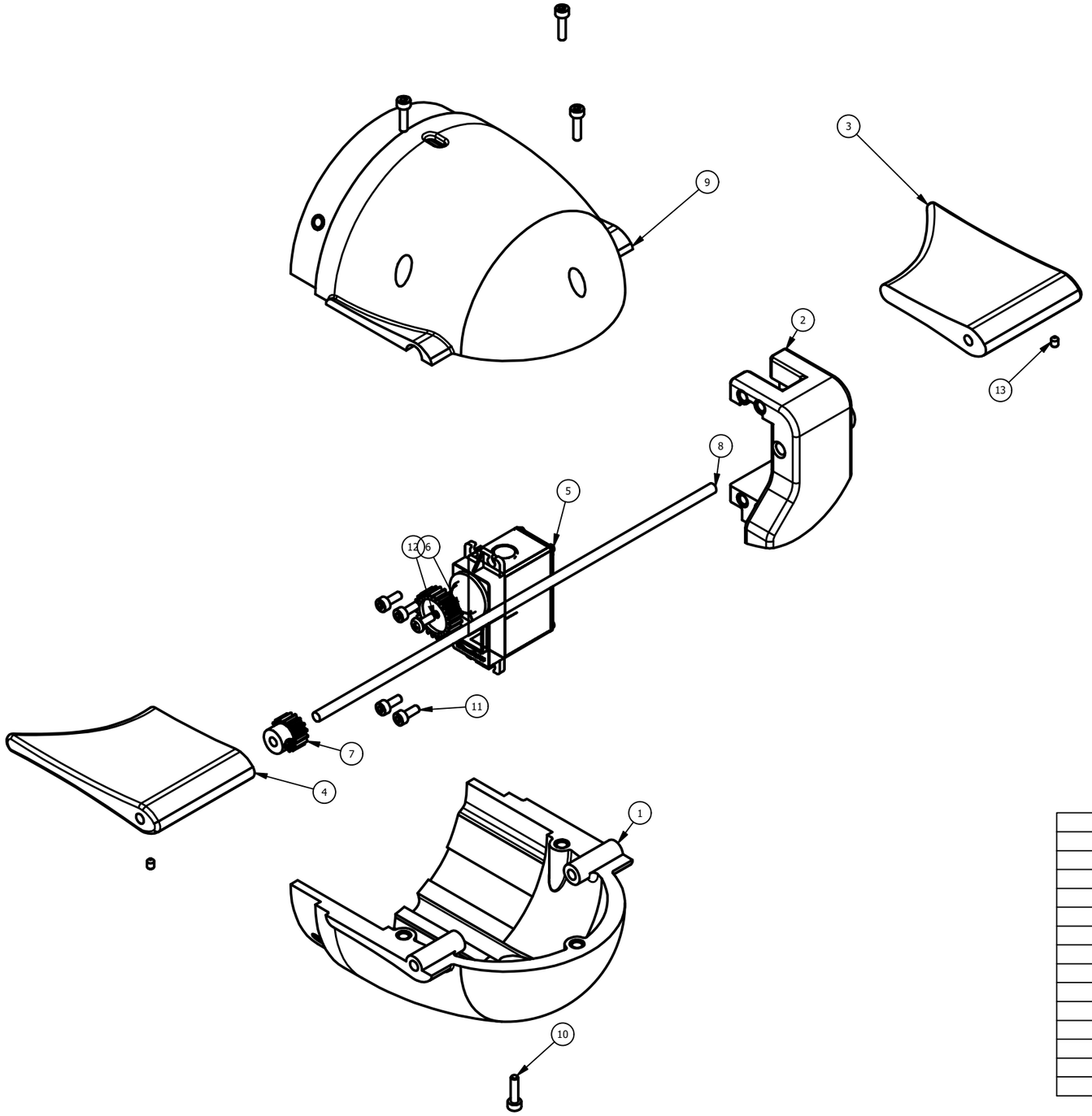
THE HEIGHTS SCHOOL - TEAM ORION

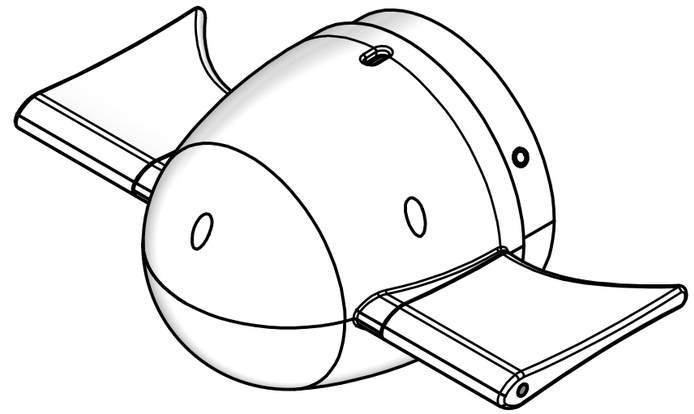
DOCUMENT NAME

AFT SERVO MOUNT

SCALE 1 : 1	PART NAME Custom Servo Holder V3		29/06/2023
SIZE A3	SHEET 17 OF 27		REF NO.

DO NOT SCALE





SHADED ISOMETRIC (1 : 2)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	Custom Fore Cap R1	
2	1	Custom Fore Servo Mount R1	
3	1	Custom Short Control Plane V6	
4	1	Custom Short Control Plane Starboard V6	
5	1	Servo Assembly	
6	1	Fore Servo Gear	STEP AP203
7	1	4mm bore gear for fore rod	STEP AP203
8	1	Custom For Cap Rod	
9	1	Custom Fore Cap Top R1	
10	4	AS 1420 - 1973 - M3 x 12	ISO metric hexagon socket head cap screws
11	4	AS 1420 - 1973 - M3 x 8	ISO metric hexagon socket head cap screws
12	1	AS 1427 - M3 x 8	Pozidriv ISO metric machine screws
13	2	AS 1421 - M3 x 4 Cup Point	ISO metric hexagon socket set screws

NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES

GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	ORION
REV	6
MATERIAL	VARIES

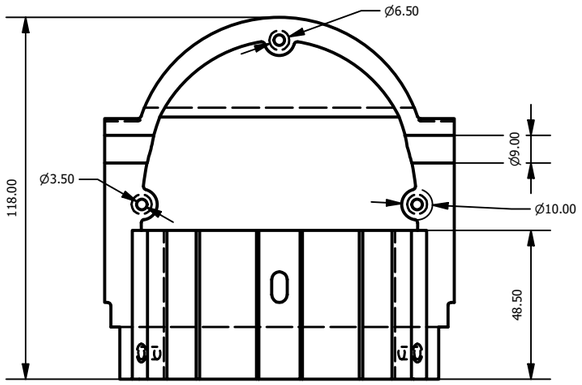
THE HEIGHTS SCHOOL - TEAM ORION

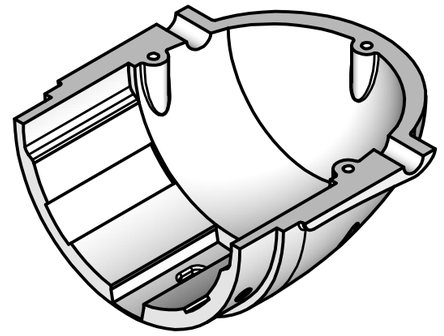
DOCUMENT NAME

FORE CAP

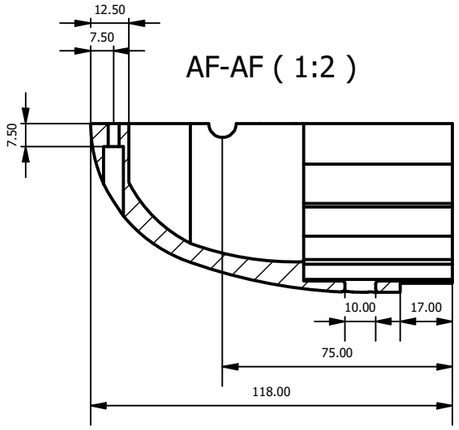
SCALE 5:10	PART NAME Custom Fore Cap Assembly V6.ipt		21/04/2023
SIZE A3	SHEET 18 OF 27		REF NO.

DO NOT SCALE

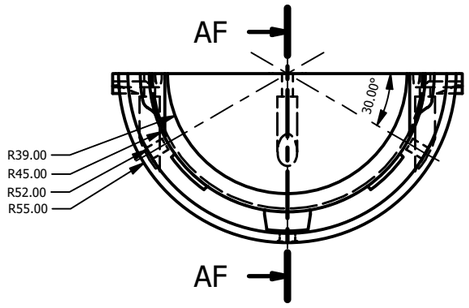


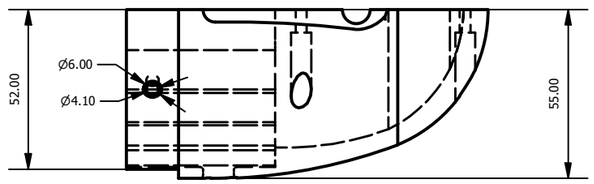


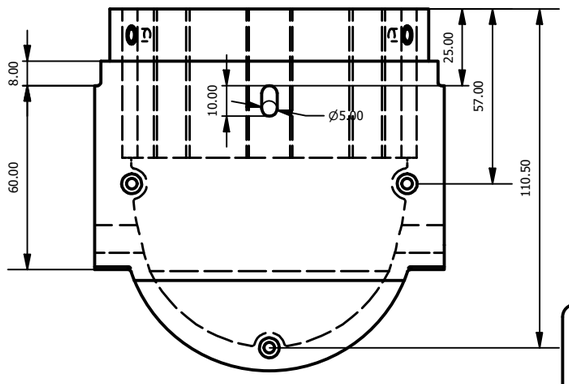
SHADED ISOMETRIC (1:2)



AF-AF (1:2)







NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	ORION
REV	2
MATERIAL	ASA PLASTIC

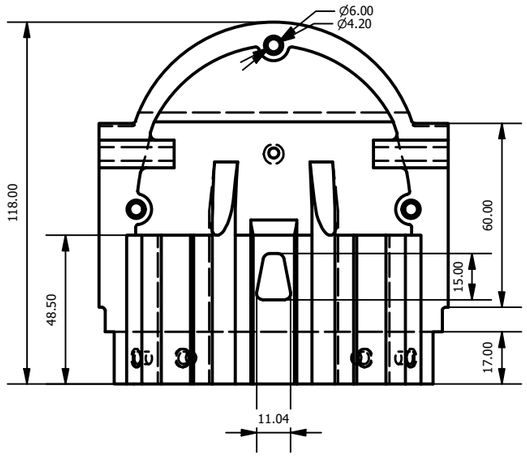
THE HEIGHTS SCHOOL - TEAM ORION

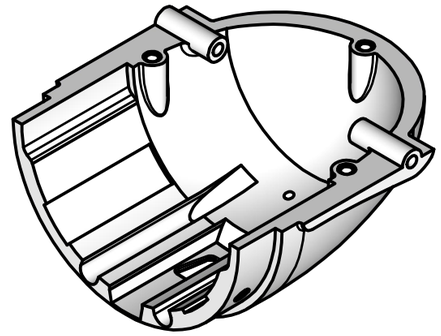
DOCUMENT NAME

FORE CAP TOP

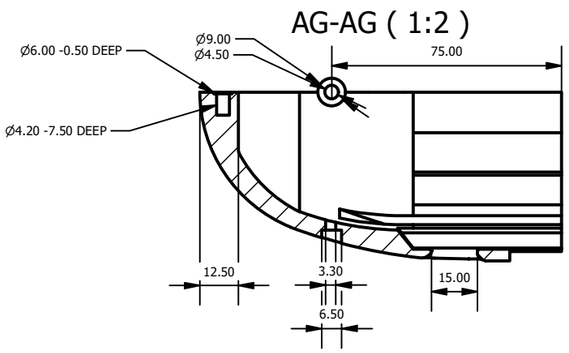
SCALE 1:2	PART NAME Custom Fore Cap Top R1		23/04/2023
SIZE A3	SHEET 19 OF 27		REF NO.

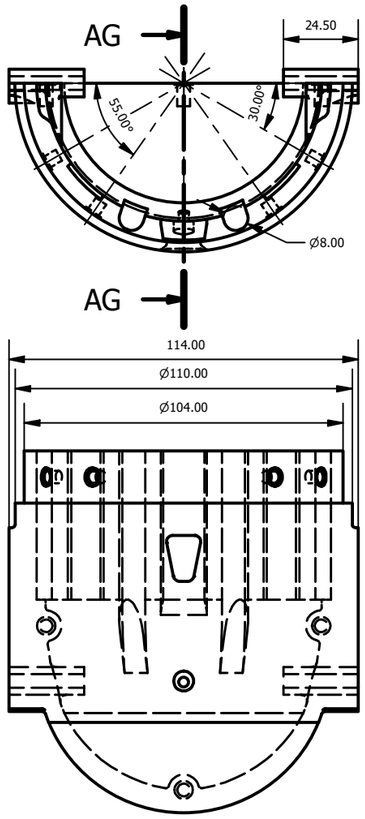
DO NOT SCALE

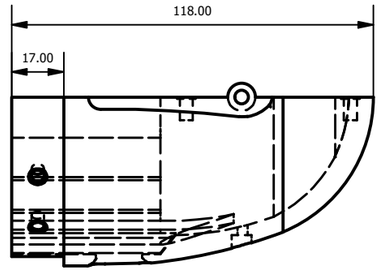




SHADED ISOMETRIC (1:2)







NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	ORION
REV	2
MATERIAL	ASA PLASTIC

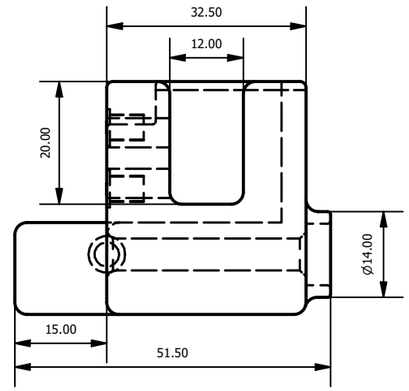
THE HEIGHTS SCHOOL - TEAM ORION

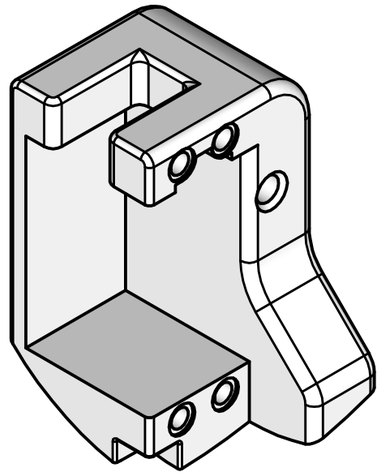
DOCUMENT NAME

FORE CAP BOTTOM

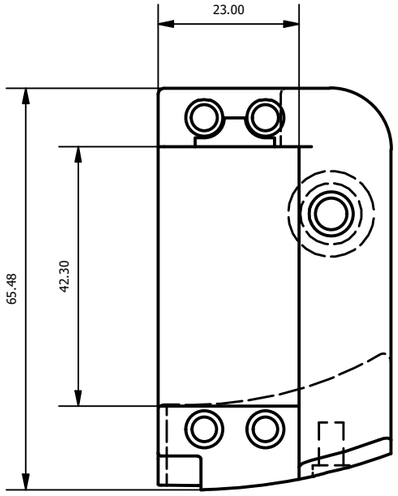
SCALE 1:2	PART NAME Custom Fore Cap R1		15/04/2023
SIZE A3	SHEET 20 OF 27		REF NO.

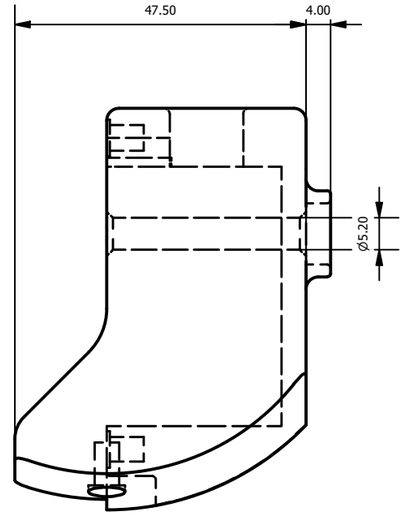
DO NOT SCALE

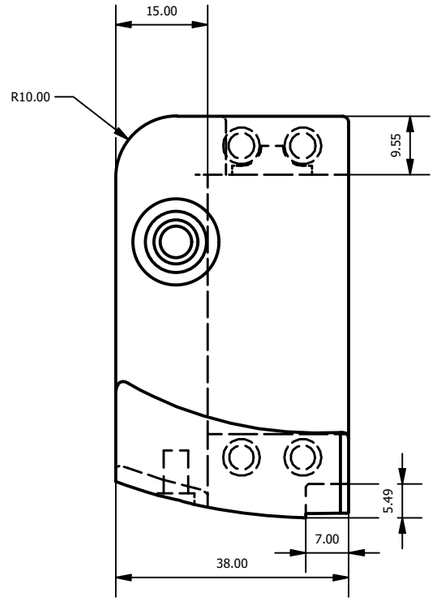


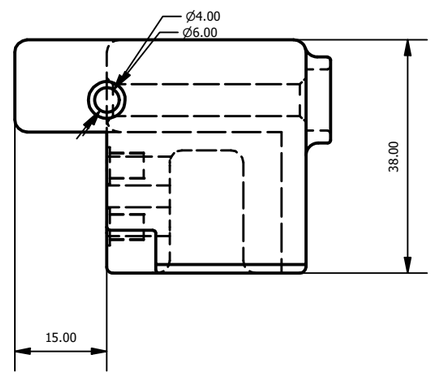


SHADED ISOMETRIC (1:1)









NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	ORION
REV	2
MATERIAL	ASA PLASTIC

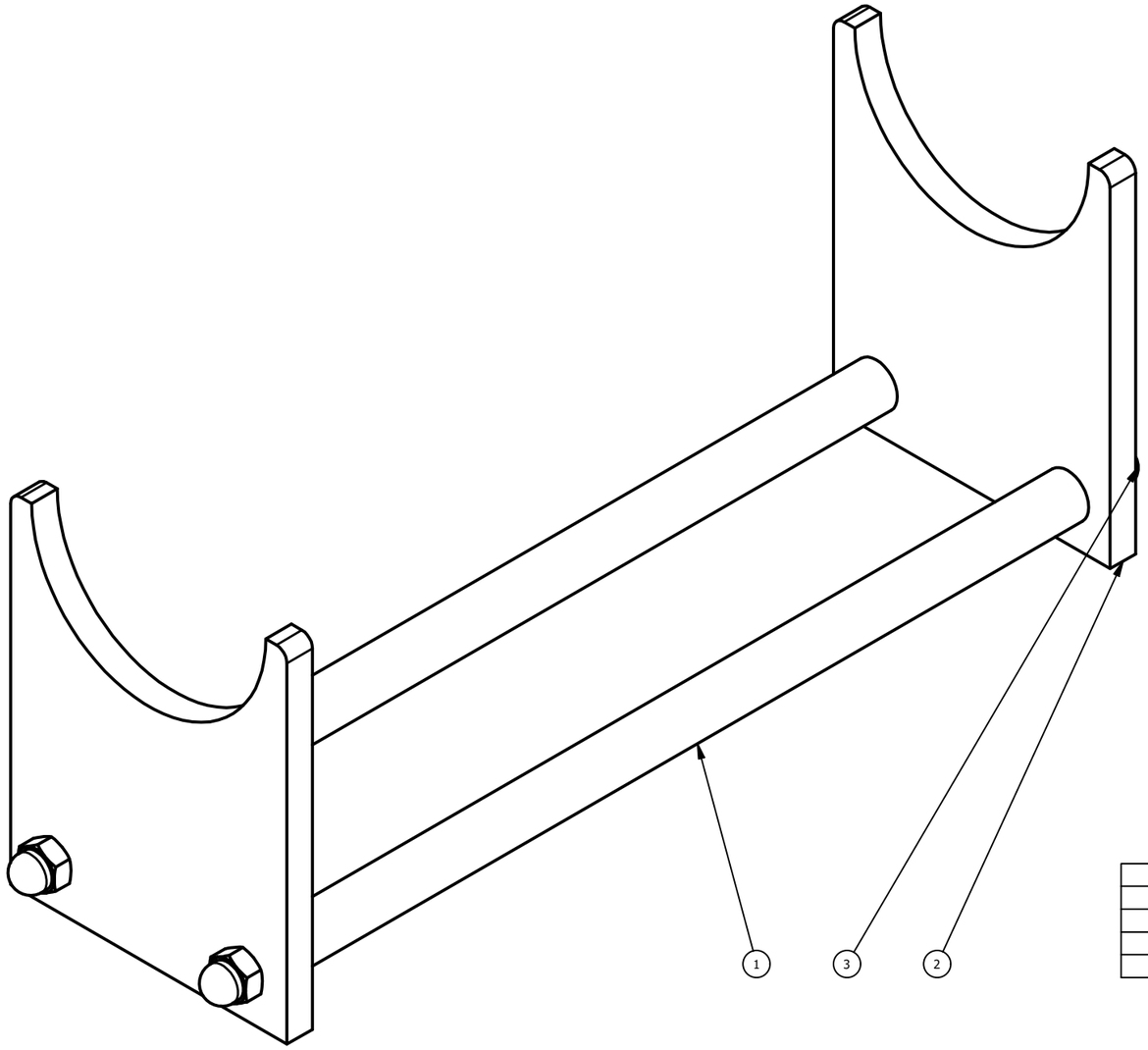
THE HEIGHTS SCHOOL - TEAM ORION

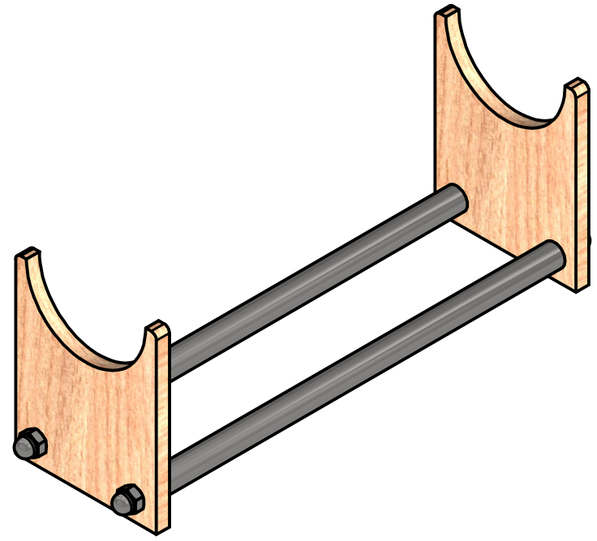
DOCUMENT NAME

FORE SERVO MOUNT

SCALE 1:1	PART NAME Custom Fore Servo Mount R1		21/04/2023
SIZE A3	SHEET 21 OF 27		REF NO.

DO NOT SCALE





SHADED ISOMETRIC (1 : 4)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	2	Sub Stand Rod	
2	2	Sub stand Sides	
3	4	JIS B 1183 Type 1 - M10x1.25	Hexagon Domed Cap Nuts

NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS
UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS
UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	Orion
MFR	Orion
REV	1
MATERIAL	Various

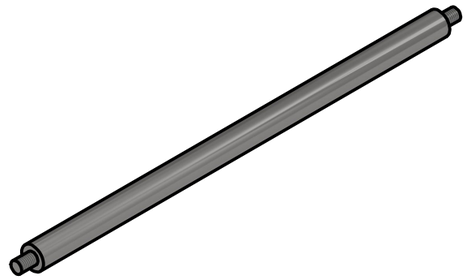
THE HEIGHTS SCHOOL - TEAM ORION

DOCUMENT NAME

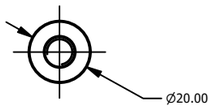
Submarine Stand

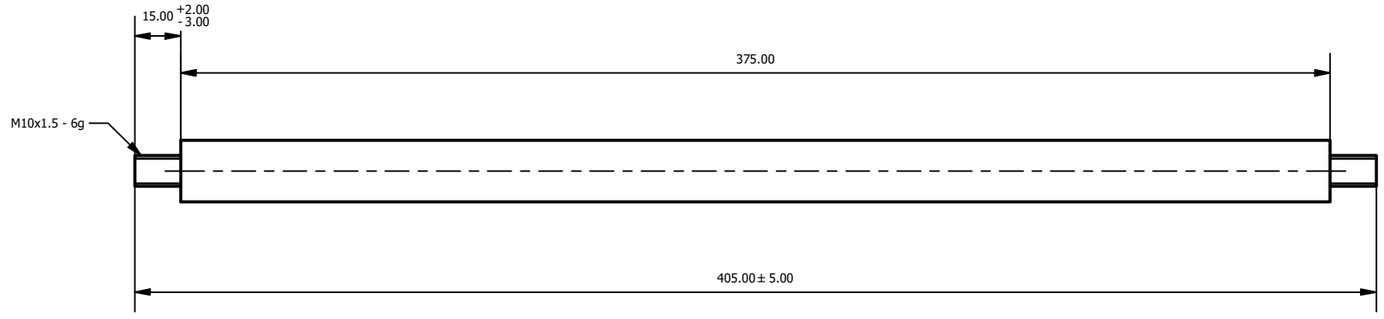
SCALE 1 : 2	PART NAME Sub Stand V2		30/07/2023
SIZE A3	SHEET 22 OF 27		REF NO.

DO NOT SCALE



SHADED ISOMETRIC (1 : 4)





NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

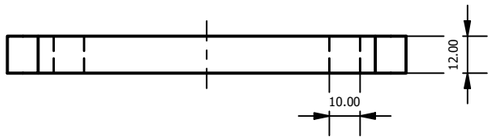
DRN	ORION
DSGN	ORION
MFR	ORION
REV	1
MATERIAL	STAINLESS STEEL

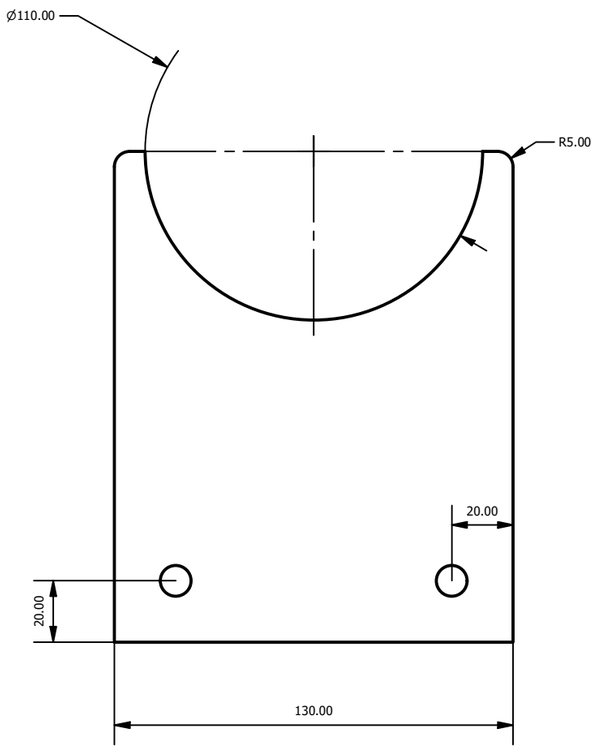
THE HEIGHTS SCHOOL - TEAM ORION

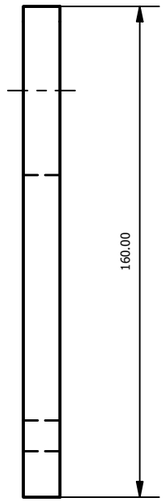
DOCUMENT NAME
SUBMARINE STAND ROD

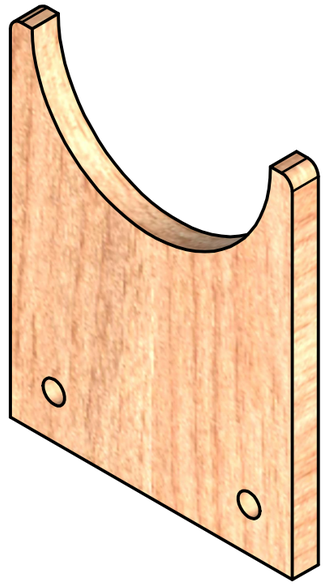
SCALE 1 : 4	PART NAME Sub Stand Rod		29/07/2023
SIZE A3	SHEET 23 OF 27		REF NO.

DO NOT SCALE





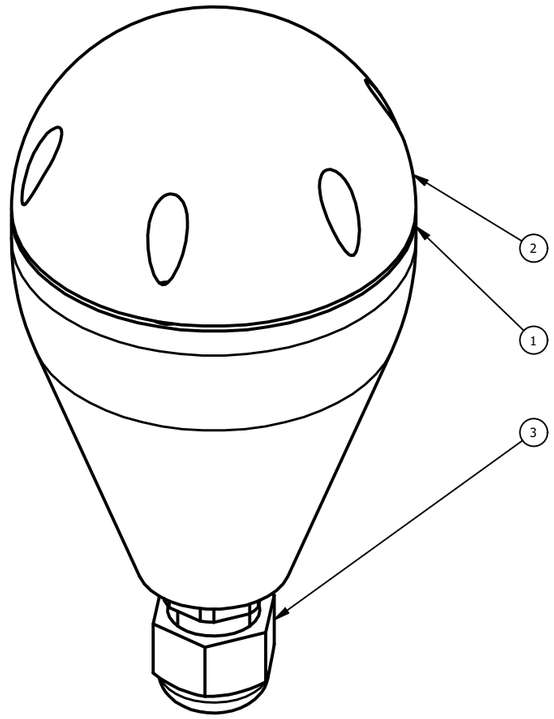


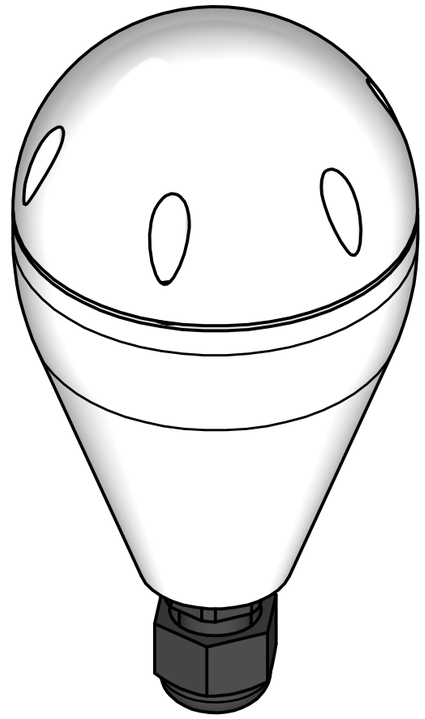


SHADED ISOMETRIC (1 : 2)

NOTES	3RD ANGLE PROJECTION	DRN ORION	THE HEIGHTS SCHOOL - TEAM ORION
		DSGN ORION	DOCUMENT NAME: SUBMARINE STAND SIDES
		MFR ORION	
		REV 1	
	DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE	MATERIAL MDF 12MM	
	TOLERANCES: GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE: ±0.02MM ±0.01°	ORION SUBS in Schools Australia THE HEIGHTS SCHOOL Preschool - Year 12	SCALE 1 : 2 · PART NAME: Sub stand Sides · 29/07/2023
			SIZE A3 · SHEET 24 OF 27 · REF NO.

DO NOT SCALE





SHADED ISOMTERIC (1 : 1)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	body	
2	1	Top	
3	1	PG9 Gland	PG 9 CABLE GLAND

NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS
UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS
UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	ORION
REV	1
MATERIAL	ASA PLASTIC

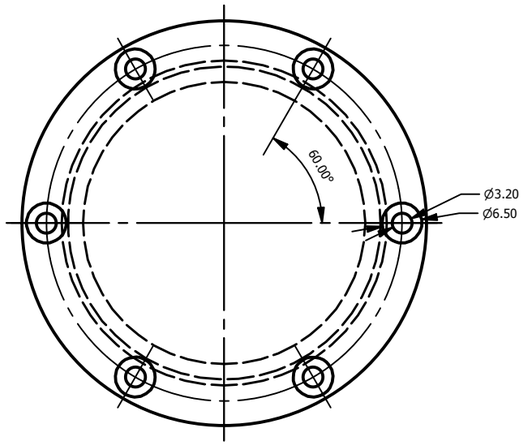
THE HEIGHTS SCHOOL - TEAM ORION

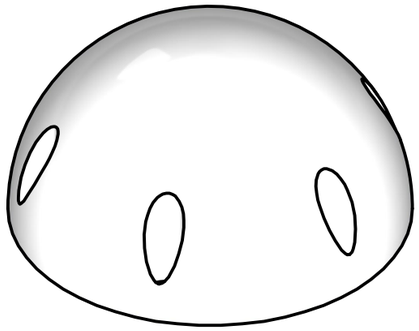
DOCUMENT NAME

BUOY ASSMEBLY

SCALE 1 : 1	PART NAME Bouy Rev 5		15/08/2023
SIZE A3	SHEET 25 OF 27		REF NO.

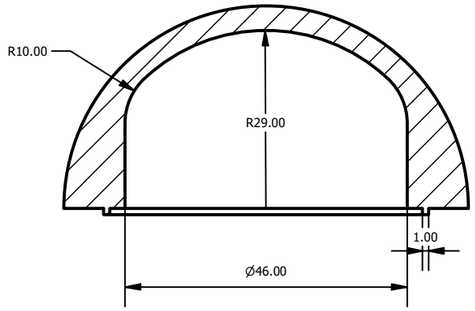
DO NOT SCALE

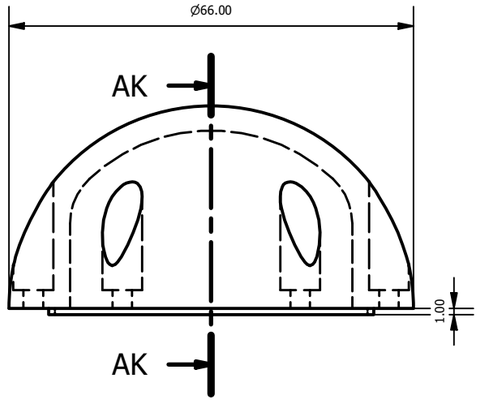


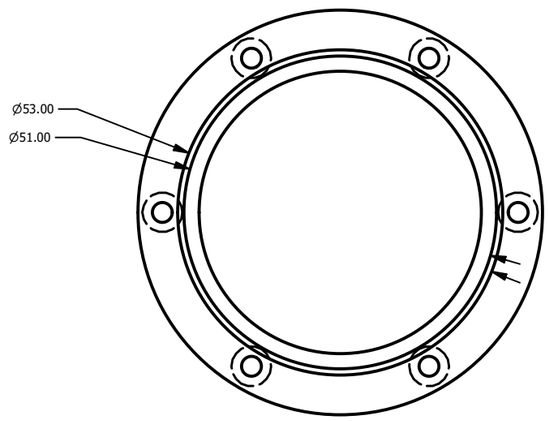


SHADED ISOMETRIC (1 : 1)

AK-AK (1 : 1)







NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	ORION
REV	1
MATERIAL	ASA PLASTIC

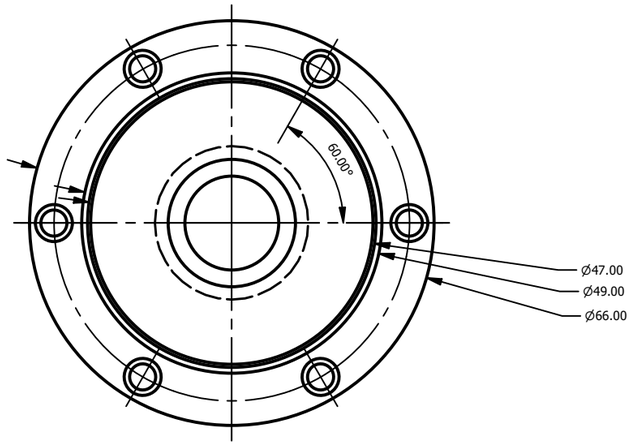
THE HEIGHTS SCHOOL - TEAM ORION

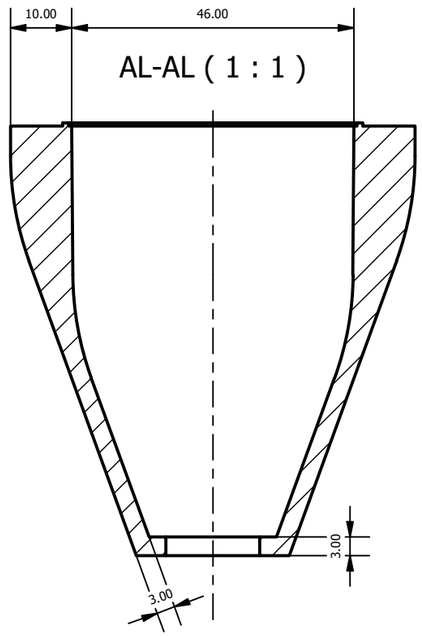
DOCUMENT NAME

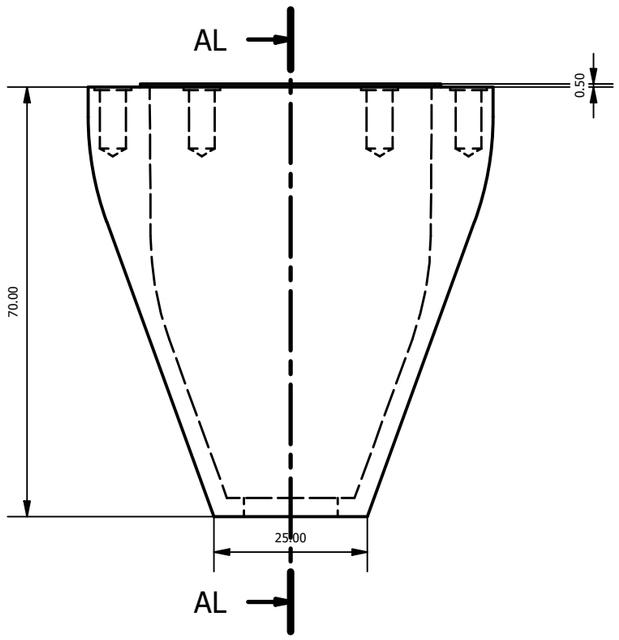
BUOY TOP

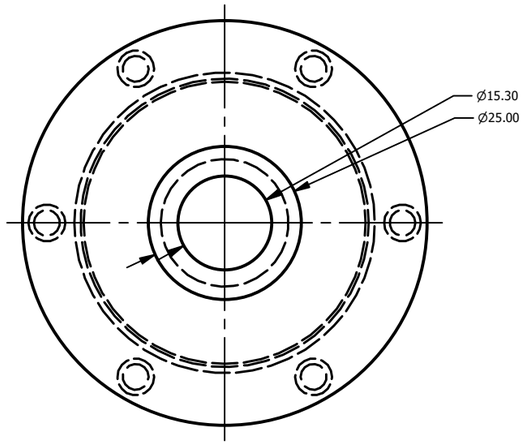
SCALE 1 : 1	PART NAME Top		15/08/2023
SIZE A3	SHEET 26 OF 27		REF NO.

DO NOT SCALE











SHADED ISOMETRIC (1 : 1)

NOTES

3RD ANGLE PROJECTION

DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE

TOLERANCES
GENERAL TOLERANCES ARE AS FOLLOWS UNLESS SPECIFIED OTHERWISE:
±0.02MM ±0.01°

DRN	ORION
DSGN	ORION
MFR	ORION
REV	1
MATERIAL	ASA PLASTIC

THE HEIGHTS SCHOOL - TEAM ORION

DOCUMENT NAME

BUOY BOTTOM

SCALE 1 : 1	PART NAME body		15/08/2023
SIZE A3	SHEET 27 OF 27		REF NO.